

Basic income versus fairness: redistribution with inactive agents

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Abstract

I study redistribution between inactive, unemployed, and employed agents who are heterogeneous in their preferences and their productive skills in the labor market and at home. The government considers that redistribution across skills is desirable, but redistribution across preferences is not: if everyone has the same preferences, equality is optimal, but if everyone has the same skills, *laissez-faire* is optimal. In turn, this leads to a simple sufficient statistic to assess any reform: any extra dollar of transfer should go to the agent whose consumption is the furthest below what she would have consumed if everyone had average skills. In high-income countries, the gap between the lowest consumption and the average productivity is empirically much larger in the labor market than among inactives. Hence, introducing an inactivity benefit, or basic income, would neither be welfare-improving nor optimal for that social welfare function, regardless of labor supply elasticities.

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1 Introduction

By law, voluntarily inactive agents are typically ineligible for social assistance programs because these transfers are conditioned on labor market participation in developed countries.¹ However, in recent years, this standard scheme has been criticized by those advocating for the introduction of a universal basic income, which is a social benefit granted on an individual basis without a means test or any work requirements.² In particular, introducing a basic income would amount to granting some positive transfers to able-bodied inactives, i.e. an inactivity benefit.³

As with any transfers, a basic income may bring equity gains by reducing the welfare inequality between actives and inactives, but it comes with some efficiency costs as it disincentivizes job-seeking efforts of unemployed agents as well as work effort provided by employed individuals. While the equity gains of a basic income have not received much attention, its efficiency costs are subject to considerable disagreement in the literature: labor supply elasticities and the associated efficiency costs of transfers are found to be large in Conesa et al. (2023), Daruich and Fernández (2024), and Golosov et al. (2024) but small in Cesarini et al. (2017). In this paper, I measure the equity gains of introducing a basic income *whatever the size of its efficiency costs*.

In the model, agents decide whether to be active or inactive, and how many hours to work when active. Most importantly, they differ in their productivity in labor market and at home as well as in their preferences. This allows us to capture relevant features of the policy question, because a government transfer will have redistributive consequences not only across different skills, as in the standard Mirrlees (1971) model, but also across agents with heterogeneous disutility of participation and preference for leisure. Yet, these multiple dimensions of heterogeneity bring a serious difficulty: how can we compare the welfare of agents with heterogeneous tastes and aggregate them in a social welfare function? In other words, how can we measure equity gains?

I build social welfare based on three key axioms. First, I impose the standard Weak Pareto principle. Second, the government does not wish to redistribute across heterogeneous preferences: if everyone has the same skills, the optimal policy is *laissez-faire*, i.e. the absence of redistribution, even if there remain consumption inequalities due to unequal preferences. Third, the government wishes to redistribute across skills: redis-

¹This is the case for all social assistance programs for the 29 developed economies studied in this paper. An overview of the legislation for European countries can be found in the MISSOC (2021) database. We discuss enforcement issues later in the paper.

²Academic supporters include notably Atkinson (1996) and Van Parijs and Vanderborght (2017) while the policy movement includes the Basic Income Earth Network (BIEN) (<https://basicincome.org/faqs/>).

³Because of this equivalence, the paper uses the terms ‘basic income’ and ‘inactivity benefit’ interchangeably. Inactivity must be understood as voluntary nonparticipation throughout the paper, except in Appendix C where I explore involuntary nonparticipation driven by circumstances.

tributive transfers between agents with identical preferences but unequal skills improve social welfare. This compensation (for one's skills)-responsibility (for one's preferences) approach, or *equality of opportunity*, has been pioneered by Fleurbaey and Maniquet (2006, 2007, 2011, 2018) for the standard income tax problem.

Because the government considers that redistribution across skills is desirable but redistribution across preferences is not, it will ask a counterfactual question: how much would each agent consume if inequalities in skills were eliminated and everyone had the *average* productivity in the labor market and at home? The difference between that benchmark and her actual consumption, or money-metric utility, measures how much of her current situation reflects bad luck in skills rather than a choice due to preferences. Because of inequality aversion in skills, the government wishes to reduce this difference, which leads to a social welfare function that *maximins* a money-metric utility.

When applied to the redistribution problem between actives and inactives, this social welfare function reduces to a sharp and simple criterion. The government will compute the gap between the worst consumption at home and the average home production, and compare it to the gap between the worst consumption in the labor market and the average wage. The main theorem of the paper shows that any welfare-improving reform consists in reducing the largest of these two gaps. As a consequence, the government should increase the inactivity benefit (resp. social assistance) whenever the difference between the worst consumption and the average production among inactives (resp. actives) is larger than among actives (resp. inactives).

This implies that the desirability of an inactivity benefit does not depend on the consumption inequalities *between* actives and inactives, but rather *among* actives and *among* inactives. Intuitively, this comes from the fact that, once inequalities *among* actives and *among* inactives vanish, all residual consumption inequalities *between* them are entirely due to unequal preferences and, therefore, are unproblematic for the government.

Because the theorem holds regardless of the government's budget constraint and for any labor supply elasticities, it is only a statement about the desirability of a reform, not its fiscal feasibility. Hence, this theorem amounts to a sufficient statistic to assess the equity gains of any tax-benefit reform. Conveniently, this sufficient statistic holds whether the current tax system is optimal or not⁴, whether the tax reforms under consideration are local or global, and whatever the empirical correlations between skills and preferences.

⁴Second-best optimum might be unreachable for real-world tax-benefit systems because of political economy constraints (Bierbrauer et al., 2021), or more generally because actual governments do not behave like the Mirrleesian planner (Stantcheva, 2016). Hence, policy recommendations of practical use are more likely to emerge from the study of welfare-improving reforms.

Once we know how to find welfare-improving reforms, we can also study optimal policies. The paper derives a partial characterization of optimal transfers in this environment with multidimensional heterogeneity. In particular, I show that they should be designed such that the gap between the worst consumption and the average production among actives is equal to the gap among inactives.⁵ This uniquely pins down the relative level of transfers between actives and inactives, that is the level of optimal inactivity benefit for a given level of optimal social assistance.⁶ In turn, if optimal social assistance increases by a dollar and the gap among actives shrinks by a dollar, then the optimal inactivity benefit should also increase by a dollar to restore the equal-gap condition. This suggests that basic income should supplement rather than crowd out existing transfer programs.

Would the introduction of a basic income be desirable? It is ultimately an empirical question. In the last part of the paper, I study whether a dollar of inactivity benefit would be welfare-improving or optimal in 29 high-income countries. To compute the sufficient statistics, one only needs the average wage, the level of social assistance (or other safety net programs), and the gap between the lowest and average home production. I elicit conservative bounds on home production by exploiting data from the *Global Survey on Working Arrangements* (G-SWA) on time savings when working from home (Aksoy et al., 2023). I combine these estimates with data on current tax-benefit systems for childless singles and lone parents (OECD, 2020).

Even under a series of conservative assumptions, the empirical application finds that the worst inactive is closer to the average home production than the worst active is to the average wage. This holds whether I consider the worst consumption when active to be a full-time minimum wage worker or a social assistance recipient. As a result, all 29 governments should first increase transfers to actives before any dollar spent on an inactivity benefit constitutes a welfare improvement. This is equivalent to saying that introducing a basic income would not be welfare-improving in all those countries.

Next, I quantify lower bounds on these increases in transfers to actives to justify any dollar of basic income. I find that these magnitudes are almost always unrealistically large: on average, governments should at least triple the level of social assistance before any dollar of basic income is welfare-improving. In sum, the inquiry shows that either the overall amount of social transfers is much too low in developed economies, or granting an inactivity benefit cannot be welfare-improving for that social welfare function.

⁵By contradiction, if the gap, say, in the labor market was larger than at home at the optimum, then a cut of one dollar in inactivity benefit combined with an increase of a dollar in social assistance would be welfare-improving, which contradicts optimality.

⁶Their joint absolute value can only be obtained by taking a stance on the efficiency costs of transfers and adding a government budget constraint.

Overall, this paper suggests that the equity gains of granting some benefits to voluntarily inactive agents are tenuous. This demonstrates a normative tension⁷ between basic income⁸ and equality of opportunity. This holds under a series of conservative assumptions. In particular, the government does not have a preference for labor market production over home production. This also holds independently of the efficiency costs of transfers, that is whether labor supply elasticities are large or small and for any revenue requirement.

Literature

Assessing the desirability of basic income with respect to equality of opportunity is particularly relevant for several reasons⁹. First, recent surveys have suggested that equality of opportunity has received some public support (Saez & Stantcheva, 2016; Stantcheva, 2021; Weinzierl, 2017). Second, these axioms are rooted in a long tradition in political philosophy (Dworkin, 1981; Fleurbaey, 2008; Rawls, 1971; Van Parijs, 1995, 2021). Third and most importantly, philosophers sharing closely related ethical standpoints disagree on the policy recommendations that it entails¹⁰. Rawls (1988) argued that his difference principle does not justify that *Malibu surfers should be fed* as they enjoy so much leisure that they are not among the worst-off. Van Parijs (1991) countered that one should not take a stance on what a good life is, and granting a basic income would allow anyone to enjoy as much leisure as one wishes, including the worst-off, thereby providing Rawlsian justification to basic income.

The study of conditionality of welfare benefits has a long history in economics¹¹ and basic income has been recently studied by e.g. Banerjee et al. (2019), Conesa et al. (2023), Daruich and Fernández (2024), Ghatak and Maniquet (2019), Golosov et al. (2024), and Hoynes and Rothstein (2019). However, most papers consider utilitarian welfare or do not exploit heterogeneous tastes. The main contribution of the present

⁷However, this tension is not an impossibility. In particular, holding agents responsible for their preferences may be normatively questionable in light of the literature on behavioral welfare economics (Bernheim, 2009; Bernheim & Taubinsky, 2018). Appendix C considers that some agents suffer from a stigma cost of participating in the labor market (as in Besley and Coate (1992a) and Moffitt (1983)) that the government compensates for. I show that if governments are ready to pay on average three times their current safety net in order to compensate for the welfare recipient stigma (or any other circumstances-driven disutility of participation), then introducing a basic income may be a welfare-improving reform.

⁸I only study the desirability of the conditionality of social benefits to labor market participation, while basic income proposals additionally require waiving conditionalities to means and to the household composition (Van Parijs, 1995). Arguably, the desirability of an inactivity benefit is a first step for the study of the desirability of a fully fledged basic income.

⁹Let me note that I do not aim to endorse a single view of social welfare but rather to link a practical policy recommendation with transparent ethical underpinnings.

¹⁰Notoriously, neither Rawls nor Dworkin endorsed themselves the view that an inactivity benefit would be justified by their theories of justice. As Dworkin (2000) puts it "*forced transfers from the ant to the grasshopper are inherently unfair*" (p.329).

¹¹See Besley and Coate (1992b, 1995), Boadway and Cuff (2014), Boadway et al. (2003), and Boone and Bovenberg (2013) among many others.

paper is to derive a simple and estimable sufficient statistic to guide policy-making even under multidimensional heterogeneity in preferences, home production and wages. This is used to characterize the difference of transfers that should hold between unemployed and inactives.

The paper also contributes to optimal taxation theory. In the canonical Mirrlees (1971) model, agents may only react to tax-benefit reforms by decreasing their hours worked i.e. on the intensive margin. This class of models has been amended to allow for participation decisions in Choné and Laroque (2005, 2011), Diamond (1980), Hansen (2021), Jacquet et al. (2013), and Saez (2001, 2002). However in these *pure* extensive margins models, there is no difference between an unemployed and an inactive. Several papers have then added search frictions to rationalize involuntary unemployment together with endogenous participation decisions (Hungerbühler & Lehmann, 2009; Hungerbühler et al., 2006; Jacquet et al., 2014; Lehmann et al., 2011). Yet, they do not allow the government to distinguish the transfers it gives to the inactive from those to the unemployed.

Boadway and Cuff (2018) allow the government to differentiate the transfers to nonparticipants from the transfers to the (involuntary) unemployed. Nonetheless, their model assumes homogeneous preferences, no intensive margin and a piece-wise linear income tax schedule while the present paper relaxes all these three assumptions. Kroft et al. (2020) also operate this differentiation and do not have any of the aforementioned shortcomings. Let me pinpoint three main differences with that paper. First, they model wages as endogenously determined in general equilibrium while in the present paper, wages are left exogenous. Second, they have a (Bergson-Samuelson) weighted utilitarian social welfare function whereas I consider an (Arrowian) social ordering function that reflects the ethics of equality of opportunity. Third, they focus on the derivation of the optimal tax system but I will also look at welfare-improving reforms, even in a suboptimal world. As I model a home sector¹² and a formal sector, the paper is related to the Mirrleesian optimal tax derivation of Rothschild and Scheuer (2013) in the multi-sector Roy (1951) model. The important difference is that they assumed that the tax schedule is uniform across sectors while I do not. In particular, it will be assumed that outcomes from the home sector are unobservable, and that the government can only give a lump-sum amount to all inactives. Beaudry et al. (2009) also assume that home production is unobservable but hours worked are observed and they focus on the difference between social assistance and unemployment insurance.¹³ In contrast here, I

¹²Gayle and Shephard (2019) introduced home production in an optimal income tax model. However, their focus is different as they estimate a large structural microeconomic model, with a marriage market and they focused on the jointness of spouses taxation.

¹³For a more recent treatment of the redistribution versus insurance problem in cash transfers and unemployment insurance, see Ferey (2022).

focus on the difference between social assistance and inactivity and assume that gross incomes are observed along with the activity binary decision.

The observability of the binary activity decision supposes that there exists a screening mechanism that the government uses to differentiate inactives from actives. In the main body, this screening mechanism is costless and perfect but we discuss the relevance of the findings for the real world where that device is costly and fuzzy in Sections 5.1 and 6. The question of the design of ordeals is left aside and has been treated by e.g. Kleven and Kopczuk (2011) and Rafkin et al. (2024).

I outline two additional differences with standard optimal tax papers. First, the literature since Saez (2001, 2002) typically studies small local tax reforms. By contrast, this paper derives a sufficient statistic that can assess any tax reform, including (potentially suboptimal) large and global ones such as the introduction of a basic income. Second, this paper does not need to impose any structural assumption on preferences¹⁴ nor any correlation between heterogeneity dimensions, while optimal tax papers typically do so to solve the multidimensional screening problem. Both facts are possible thanks to the axiomatic derivation of a social objective as a transitive ordering between any two allocations i.e. any two tax-benefit systems. This endeavor has been inspired by the fair income tax literature (Fleurbaey & Maniquet, 2006, 2007, 2011, 2018). The present paper also contributes to the latter by including inactive agents as well as additional dimensions of heterogeneity and deriving the axiomatic characterization in this new environment.

In section 2, I formalize the environment. In section 3, I construct the social objective in the first-best. In section 4, I introduce the second-best environment where the government cannot observe skills and preferences but can tag transfers to income and labor market participation. I derive the main theoretical results. In section 5, I present the empirical application. In section 6, I conclude.

2 Model

There is a finite set $\mathcal{I} = \{1, 2, \dots, I\}$ of agents whose generic element is denoted by i . There are only two goods, consumption and labor, forming a bundle denoted by $(c_i, l_i) \in X$. The homogeneous consumption good $c_i \in \mathbb{R}_+$ is produced either in the home sector or in the labor market. The labor supply variable l_i is set to -1 when the agent stays at home, or takes value of hours worked in a normalized interval $[0, 1]$ when the agent is in the labor market. Hence, an inactive agent has $l_i = -1$, an unemployed

¹⁴Dworczak (2026) mechanism design exercise finds a sufficient condition for the transfers to those meeting ordeals to be larger than transfers to those not meeting ordeals if preferences are quasilinear. In the present paper, there are no quasilinear restrictions. Moreover, the optimal transfers exercise precisely quantifies the optimal difference of transfers between inactives and unemployed.

agent has $l_i = 0$ and an employed agent has $l_i \in (0, 1]$.

Each agent is endowed with a convex preference ordering $\succsim_i$ that can be represented by a continuous ordinal utility function $u_i(c_i, l_i)$ which is strictly increasing in c_i and non-increasing in l_i . An agent may have an idiosyncratic disutility of participation¹⁵ to the labor market whenever $u_i(c_i, -1) - u_i(c_i, 0) \geq 0$ which captures the utility loss for an inactive that becomes unemployed while keeping the same level of consumption. Importantly, the disutility of participation must be distinguished from the willingness to work.¹⁶ In this framework, disutility of participation embodies a preference to produce at home rather than in the labor market, while willingness to work reflects the substitutability between consumption and hours worked on the labor market.

In addition to their preferences, agents are also heterogeneous along their vector of innate productive abilities $(w_i, h_i) \in [\underline{w}, \bar{w}] \times [\underline{h}, \bar{h}] \subseteq \mathbb{R}_+^2$, where the first coordinate denotes the marginal productivity used on the labor market and the second coordinate captures the *surplus* of home production that inactivity allows for relative to activity, not its *level*.

In the labor market, I retain the standard assumption of a constant return to scale technology whose sole input is hours worked, and its marginal productivity is given by w_i . In the home sector, h_i captures the surpluses of production that active agents lose by joining the labor market. It is a reduced-form term for outcomes of activities such as gardening, child rearing or housekeeping. Obviously in reality all agents produce at home, be they inactive, unemployed or employed. Hence, it is implicitly assumed here that all actives produce an identical level at home, normalized to 0 (i.e. the first-best Laissez-faire consumption of unemployed agents) and inactives produce h_i more than them¹⁷.

In the first-best, the second fundamental welfare theorem prescribes that efficient redistribution could be achieved through lump-sum transfers. Denoting these transfers by t_i , the first-best budgets, illustrated in [Figure 1](#), are defined by :

$$B(t_i, w_i, h_i) = \left\{ (c_i, l_i) \in X : c_i \leq a(l_i)w_i l_i + (1 - a(l_i))h_i + t_i \right\}$$

where $a(\cdot)$ is an indicator function that takes value 1 if the agent is active (i.e. $l_i \in [0, 1]$)

¹⁵It is known that the disutility of participation displays substantial heterogeneity in the cross-section of households (Kaplan & Schulhofer-Wohl, 2018). Here, it may capture (but it is not restricted to) the stigma utility cost of welfare conditionality as in Friedrichsen et al. (2018) and Moffitt (1983), see [Appendix C](#).

¹⁶In this setup, the willingness to work may be approximated by the marginal rate of substitution over non-negative values for l . A low (resp. high) marginal rate of substitution in absolute value reflects a high (resp. low) willingness to work.

¹⁷In the empirical [section 5](#), I argue that, while absolute levels of home production are typically unobservable, reasonable bounds for home surpluses h_i may be found.

or 0 if the agent stays at home (i.e. $l_i = -1$)¹⁸.

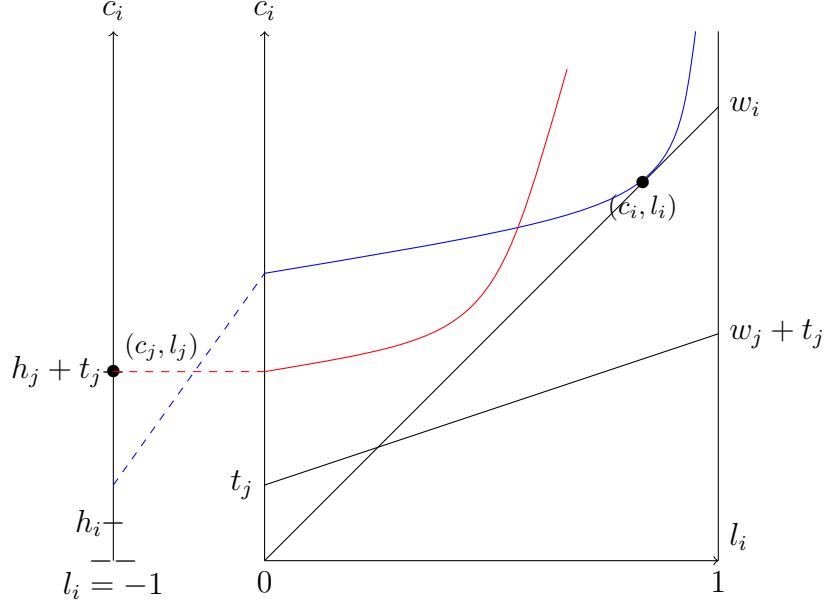


Figure 1: Illustration of the model for two agents i, j . Blue agent i receives no lump-sum transfer and has a positive disutility of participation, but still chooses optimally to be active. Contrarily, the red agent j receives $t_j > 0$ and does not display disutility of participation but chooses to be inactive.

Observe that when $a(\cdot) = 1$, this model nests the standard linear production model used e.g. in optimal taxation theory. Hence, agents can react to tax reforms along three margins in this model: (i) intensive (by decreasing hours worked when employed), (ii) standard-extensive (by switching from employed to unemployment), and (iii) participation-extensive (by switching from activity to inactivity). Four important remarks must be raised.

First, this model allows for both involuntary and voluntary unemployment. In the former case, the (primitive) state of the labor market nullifies the productivity of some agents (such that $\underline{w} = 0$) which can remain active with $l_i = 0$. By contrast, voluntary unemployment arises when $l_i = 0$ is the utility-maximizing choice of an agent endowed with some positive wage rate $w > 0$ ¹⁹.

Second, I assume that there is no intensive margin in the home sector²⁰. Of course,

¹⁸Observe that the use of this indicator function renders moot the actual value of l_i when inactive as long as it is not in $[0, 1]$. The choice of -1 is both arbitrary and harmless.

¹⁹In the first-best, unemployment (both voluntary and involuntary) only happens in the knife-edge case where $h = 0$. In the second-best, unemployment may happen when transfers to unemployed are larger than transfers to inactives, which is the empirically relevant case.

²⁰Moreover, an intensive margin in the home sector would imply that the inactivity benefit distorts effort in the home production which would be unfavorable to the emergence of a basic income. Hence, this modelling assumption is conservative with respect to my result.

in reality agents partition their time between leisure, paid work and home production. However, labor market inactivity is a binary status for any tax-benefit system, such that there is no such thing as a part-time inactive in the eyes of the fiscal authority: agents can either be full time in the home sector or not at all in this sector.

Third, inactive and unemployed agents both enjoy the full unit of leisure. However, only the former are able to produce at home. This is equivalent to saying that there is a fixed time cost spent looking for a job when unemployed that can be used productively when inactive. Hence, h_i must be understood (and measured) as the product of this time cost with the hourly idiosyncratic²¹ value of production in the home sector. For example, in a legal working week of, say, 40 hours, the inactive and the unemployed do not provide any hours worked and thus enjoy 40 hours of leisure. However, an unemployed must spend, say, 10 hours sending job applications during which the inactive takes care of his children. The inactive's h_i is then the product of $\frac{10}{40}$ with the shadow price of an hour of day care services in that economy. This estimation procedure for h_i is discussed at length in [section 5](#) below.

Fourth, the model assumes away any positive externality of job search. This is again a conservative assumption, in the sense that such externality would lay the grounds for Pigouvian subsidies to the unemployed (or Pigouvian tax on inactivity) in the redistributive problem, which would be unfavorable to the emergence of a basic income. Despite that, I find below that basic income is not desirable such that this result would be even stronger with such an externality.

Note that throughout the paper, no parametric specification of utility functions is imposed. Moreover, I will not impose any correlation between primitives $\succsim_i$, w_i and h_i , neither at the individual level nor in the cross-section. As a consequence, the results are valid for any empirical correlations observed in the data.

Finally, let me define an economy e by a list of endowments for each agent in each heterogeneity dimension $e = \{(\succsim_i, w_i, h_i)_{i \in \mathcal{I}}\}$. I denote the set of all such economies by E . An allocation is a bundle for each individual and is denoted $(c, l) = \{(c_i, l_i)_{i \in \mathcal{I}}\} \in X^{\mathcal{I}}$.

3 Fair social objective

3.1 Characterization

Each economy can yield many allocations which are associated with observable inequalities in consumption-labor outcomes (c_i, l_i) all originating from unobserved heterogeneity in primitives $(\succsim_i, w_i, h_i)$. The key question for a government is: when should an allocation

²¹In a structural macro exercise with time use data, Boerma and Karabarbounis (2021) show that (1) inequalities in the home sector are quantitatively important and (2) inequalities in home production efficiency are needed to explain the variance of home inputs conditional on wages and preferences.

tion (c, l) be socially preferred to an allocation (c', l') ?

There are infinitely many ways to answer this question. The standard approach used in economics in general and in optimal taxation theory in particular consists in using a weighted sum of utility functions to assess welfare improvements. However, when agents have heterogeneous preferences, such summation is sensitive to the cardinalization of preferences, i.e. to the choice of the particular $u_i(\cdot)$ function that represents $\succsim_i$.

There is no commonly accepted way to choose such a cardinalization. Yet, choosing it arbitrarily may lead to pathological consequences: Becko and Sztutman (2026) showed that some Pareto-efficient tax-benefit systems may only be rationalized by summing utility functions with unbounded curvature with respect to consumption.²² Moreover, any cardinalization implicitly embodies a way to compare welfare interpersonally, and as such it is ethically-loaded. The literature in normative economics has acknowledged that policy recommendations are contingent on an ethical view, and as such, it is better to make it visible rather than leaving it to the researcher's choice. But how can we link ethics with a social welfare function?

A popular method in recent years has been to rely on Saez and Stantcheva (2016)'s generalized weighted sum of utilities where weights can be a function of $(\succsim_i, w_i, h_i)$ to reflect different fairness views. While this approach is flexible and tractable, it has a major drawback: it may fail transitivity in its policy recommendation (Sher, 2024).

In the present paper, I build an Arrovian social welfare function, i.e. a function that associates to each economy a transitive ordering of allocations.²³ In particular, among the many possible social welfare functions, I will select the one that respects three key fairness principles: (1) **Weak Pareto**, (2) **Responsibility** and (3) **Weak Transfer**.

Weak Pareto ensures that a Pareto-dominated allocation will never be chosen by the planner. Yet, there are typically many points on the Pareto frontier, and the next two axioms single out the one that reflects equality of opportunity. **Responsibility** requires that in the knife-edge case where all agents have the very same productive endowments w and h , then the *Laissez-faire* allocation (i.e. the absence of transfers) is the social optimum, even if unequal $\succsim_i$ still generates inequalities. As a result, **Responsibility** does

²²Becko and Sztutman (2026) studied which tax-benefit systems are compatible with a sum of utility functions with bounded curvature. Instead, the present paper does not impose the utilitarian aggregator nor the cardinalization, but it derives a single social welfare function from axioms. The advantage of the latter is that the policy recommendation is linked to ethically-transparent axioms, but the advantage of the former is that it contains a larger set of cardinalizations. Critically, their empirical test depends on the extent of heterogeneity in taxable income elasticities, while my sufficient statistics provide a policy recommendation for any elasticities.

²³This aggregation from individual preferences to social welfare follows the Arrow (1950) tradition and constructs the social welfare function axiomatically as in the pioneering work of Fleurbaey and Maniquet (2007).

not fight inequalities spawned by unequal preferences. Conversely, **Weak Transfer** requires that if two agents have identical preferences and identical labor supply decisions (i.e. identical l_i), a transfer from the richer to the poorer agents is a social improvement. Hence, **Weak transfer** fights inequalities spawned by unequal skills. As argued above, this ethical view may be relevant for the policy debate on basic income.

I now present the social welfare function that pursues the ethics of equality of opportunity: it maximins $M_i(c_i, l_i)$. In short, $M_i(c_i, l_i)$ is a money-metric utility function that measures the well-being of agent i when she consumes the bundle (c_i, l_i) as the transfer that renders this agent indifferent between (c_i, l_i) and her preferred bundle in the budget determined by the average productive vector. Loosely speaking, the further below the individual sees herself from an average agent, the worst will be her well-being.

Definition 1. *The equality of opportunity social welfare function considers that (c, l) is socially strictly preferred to (c', l') whenever*

$$\min_{i \in \mathcal{I}} M_i(c_i, l_i) > \min_{i \in \mathcal{I}} M_i(c'_i, l'_i)$$

where $M_i(c_i, l_i)$ is a money-metric utility function defined as

$$M_i(c_i, l_i) = \min t \quad \text{such that } (c_i, l_i) \sim_i \max_{\tilde{w}} B(t, \tilde{w}, \tilde{h})$$

$$\text{where } \tilde{w} = \frac{1}{I} \sum_i w_i \quad \tilde{h} = \frac{1}{I} \sum_i h_i \quad \text{and } t \in \mathbb{R}$$

The graphical construction of $M_i(c_i, l_i)$ is illustrated in [Figure 2](#) for the case of a two-agent economy that was introduced in [Figure 1](#).

[Definition 1](#) implies that the social welfare function will focus on the agent having the smallest $M_i(c_i, l_i)$, i.e. the agent with the lowest well-being. In other words, the social objective is of the *maximin* type.

The next theorem shows that the aforementioned three axioms single out the social welfare function in [Definition 1](#) when they are combined with two technical axioms. The fourth axiom, **Mean-Preserving Separability** is reminiscent of the Separability axiom used in virtually all social welfare functions since Fleming (1952): indifferent agents should not matter in the social decision.²⁴ The last axiom, **Hansson (1973) Independence**, allows the social welfare function to use indifference curves rather than mere utility levels to escape Arrow (1950)'s impossibility. All formal definitions of axioms are relegated to [Appendix A.1](#).

²⁴The axiom used in this paper is logically weaker than the standard Separability axiom. This weakening is required to avoid an impossibility, as is proven in the Appendix A.

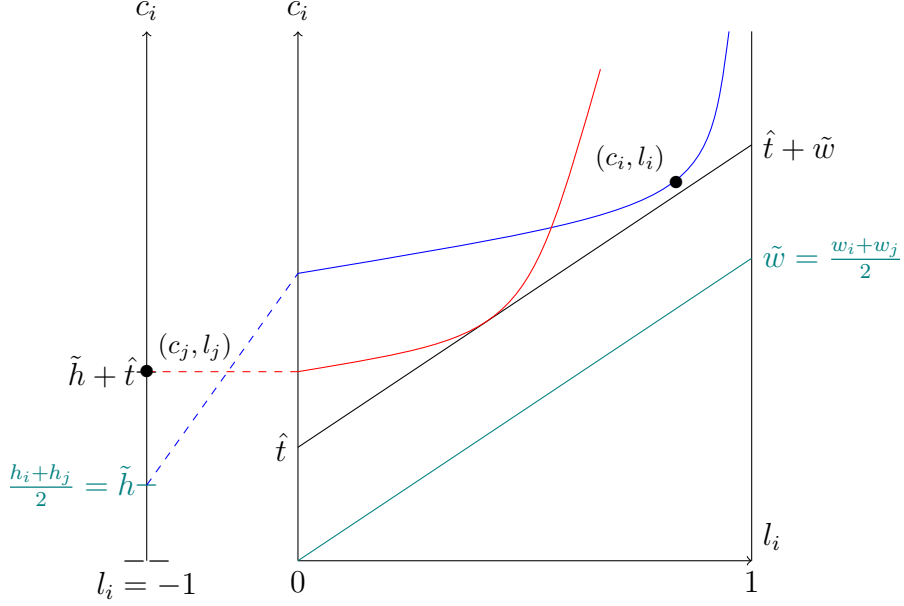


Figure 2: Construction of the money-metric welfare $M_i(c_i, l_i)$ using the example economy of Figure 1. First, one computes $(\tilde{w}, \tilde{h})$ as $(\frac{w_i + w_j}{2}, \frac{h_i + h_j}{2})$. This yields the green reference budget $B(0, \tilde{w}, \tilde{h})$. Then, one finds the welfare of an agent by computing the smallest transfer t that translates the reference budget up to tangency (or intersection) with the indifference curve. In this case, blue agent i 's indifference curve at (c_i, l_i) intersects $B(0, \tilde{w}, \tilde{h})$ when $l_i = -1$ so $M_i(c_i, l_i) = 0$, while red agent j 's indifference curve at (c_j, l_j) is tangent to $B(\hat{t}, \tilde{w}, \tilde{h})$ such that $M_j(c_j, l_j) = \hat{t} > M_i(c_i, l_i)$ and agent i is the worst-off.

Theorem 1. *If a social welfare function satisfies **Weak Pareto**, **Responsibility**, **Weak Transfer**, **Mean-Preserving Separability** and **Hansson Independence**, then it is the equality of opportunity social welfare function in [Definition 1](#).*

Proof. See [Appendix A.2](#). ■

Before providing intuition behind this result, it is useful to study its consequences for optimal policy. What would be an optimal schedule of lump-sum transfers in the first-best with respect to this social welfare function? Because of the maximin nature of the social objective, optimality in the first-best is achieved whenever all agents in the economy have an identical well-being measure, i.e. $M_i(c_i, l_i) = M_j(c_j, l_j)$ for all $i, j \in \mathcal{I}$. All agents then reach their indifference curve tangent to this reference budget set²⁵, thereby all enjoying the same level of welfare in the eyes of the planner. I illustrate a first-best optimal allocation in [Figure 3](#).

Two crucial points must be raised. First, observe that given the heterogeneity in preferences, equality of welfare at the optimum does not mean that all agents consume

²⁵This tangency condition of the optimal allocation is different from (and compatible with) the tangency condition of individuals' constrained utility maximization. For example, in [Figure 3](#), agent j optimizes by choosing (c_j, l_j) which implies that her indifference curve is tangent to her individual budget when $l_j > 0$, but it coincides with the green reference budget frontier $B(m, \tilde{w}, \tilde{h})$ when $l = -1$.

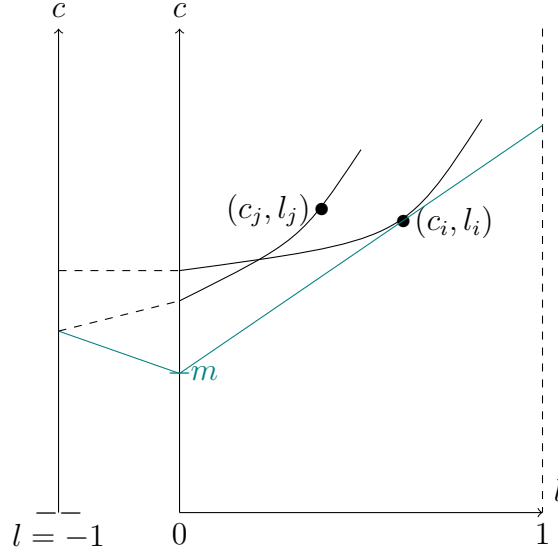


Figure 3: An allocation (c, l) equalizing $M_i(c_i, l_i) = M_j(c_j, l_j) = m$ across agents is optimal with respect to the *equality of opportunity* social welfare function in the first-best. All indifference curves are tangent (or intersect) with the green reference budget set $B(m, \tilde{w}, \tilde{h})$.

the same bundle.²⁶ Second, and most importantly, at the optimal allocation there are no productive inequalities *within* sectors, but only *between* sectors. More precisely, all indifference curves either pass through the inactive agents' consumption or are tangent to the same hourly wage over $l \in [0, 1]$. Yet, the government does not wish to redistribute further to equalize consumption across sectors, even if home production is not equal to hourly wages. This shows that a government that pursues equality of opportunity wishes to redistribute within sectors, not between sectors. Intuitively, this comes from the fact that, once within-sector inequalities vanish, the sectoral decision is entirely driven by preferences and the government does not wish to redistribute across preferences.

3.2 Why the average?

Theorem 1 asserts that the fair social objective should be (1) of the maximin type, (2) over money-metric utilities, and (3) with the particular money-metric utility function $M_i(c_i, l_i)$. If one compares these results to the literature, only the latter contains some novelty. Indeed, it is now standard in the literature on equality of opportunity that the finite inequality aversion embodied in **Weak Transfer** turns out to be infinite when combined with the other axioms.²⁷ Moreover, it comes at no surprise that money-metric utilities are a useful and flexible way to cardinalize utilities when preferences are het-

²⁶Also, note that such strongly egalitarian allocations may also be reached with a standard utilitarian setup in the first-best, as has been known since Edgeworth (1897).

²⁷See e.g. Fleurbaey and Maniquet (2011) and Piacquadio (2017) for a discussion. The key axioms leading to the maximin-type of social welfare functions are **Weak Transfer** and **Hansson (1973) Independence**. The key axiom leading to a money-metric utility is **Responsibility**.

erogeneous.²⁸ What is peculiar and new with [Theorem 1](#) is that the reference prices in the money-metric utility are pinned down by arithmetic averages $\tilde{h}$ and $\tilde{w}$. I now sketch an example to provide intuition for this result. It is illustrated in [Figure 4](#).

Consider four agents: two agents with preferences $\succsim_a$ and two with preferences $\succsim_b$. For the purpose of this example, let us assume that these are purely inelastic preferences, i.e. the labor supply choice is independent of their endowments. We know that the social ordering should be *maximinning* a money-metric utility function and wonder which reference endowments are pinned down by the axioms.

When these four agents have identical skills, the economy is $\{(\succsim_a, w, h), (\succsim_a, w, h), (\succsim_b, w, h), (\succsim_b, w, h)\}$. In such case, **Responsibility** implies that the absence of redistribution is optimal and *Laissez-faire* allocation (c^*, l^*) is optimal. Because of the maximin nature of the social objective, we must have that the money-metric welfare of all agents is equal at the optimal allocation. Which reference endowments in the economy $\{(\succsim_a, w, h), (\succsim_a, w, h), (\succsim_b, w, h), (\succsim_b, w, h)\}$ ensure that the money-metric welfare of agents a and b is equal at (c^*, l^*) ? The answer is: all those that are such that

$$\rho(\{w, h\}, \{w, h\}, \{w, h\}, \{w, h\}) = \{w, h\}$$

where ρ is the reference operator. This equation tells us that $\rho(\cdot)$ is a mean. At this stage though, it could be many means: the quadratic mean, the cubic mean and all other Hölder means respect that equation.

Now imagine that there is a shock to the endowments and the economy becomes $\{(\succsim_a, w + \delta, h + \delta), (\succsim_a, w - \delta, h - \delta), (\succsim_b, w + \delta, h + \delta), (\succsim_b, w - \delta, h - \delta)\}$ where $\delta > 0$. In other words, among each pair of agents with preferences $\succsim_a$ and $\succsim_b$, there is one winning (δ, δ) in her endowments and one losing (δ, δ) . In that case, **Weak Transfer** imposes that a within-pair transfer from the fortunate to the unfortunate is a social improvement. Hence, the allocation (c^*, l^*) constitutes a welfare improvement in the new economy relative to the absence of transfers: all δ shocks in endowments are counteracted by consumption transfers through **Weak Transfer** thanks to linear production.

Which reference vector is consistent with (c^*, l^*) being not only the outcome of welfare-improving transfers in the new economy, but also its optimum? Again, the maximin nature of the social objective requires the money-metric welfare of all agents to be equal at (c^*, l^*) . At that allocation, every agent consumes her preferred bundle in the budget $B(0, w, h)$, so all four money metrics are equal (to zero) precisely when the reference

²⁸In related papers, Bosmans et al. (2018), Fleurbaey and Maniquet (2017), and Piacquadio (2017) provided recent axiomatizations of money-metrics as arguments of social welfare functions. Their history dates back at least to Samuelson and Swamy (1974) and recent papers showed their empirical usefulness when agents have heterogeneous preferences (Baqaei et al., 2024; Jaravel & Lashkari, 2024).

4 Redistributive taxes and transfers

In this section I characterize the tax-benefit system pursuing equality of opportunity under incentive-compatibility constraints. As in Mirrlees (1971), these constraints arise because the government is unable to observe the endowment vector $(\bar{z}_i, w_i, h_i)$ of each individual.

Moreover, the government is unable to observe l_i and can only observe $y_i = a(l_i)w_i l_i$, the gross labor income reported in tax returns. I will also assume that, when $y_i = 0$, the government can perfectly distinguish the inactive from the unemployed agent. In other words, $a_i(\cdot)$ is observable even if l_i is not. This is a key assumption: it allows the government to differentiate the transfers it gives to unemployed agents from transfers to inactives. This is consistent with the observation that in most developed economies, there exists a screening mechanism enforcing the conditionality of welfare benefits to a job-seeking behavior.³⁰

For active agents with $a(l_i) = 1$, the government designs the tax schedule on the labor market through the nonlinear tax function $\tau(y)$ which is a subsidy whenever $\tau(y) < 0$ on some y . For agents out of the labor market with $a(l_i) = 0$, the government cannot observe home production yet these inactive agents may receive a subsidy $D \in \mathbb{R}$ which is a tax if $D < 0$ such that the second-best budgets are

$$B(\tau, D, w_i, h_i) = \left\{ (c_i, l_i) \in X \mid c_i \leq a(l_i)[y_i - \tau(y_i)] + (1 - a(l_i))[h_i + D] \right\}$$

I call D the inactivity benefit and $-\tau(0)$ the social assistance. Agents choose l_i to maximize $u_i(c_i, l_i)$ under this second-best budget constraint.

One may wonder about the relationship between the tax-benefit system (τ, D) studied here and the basic income proposals. Indeed, why wouldn't we give a basic income *universally* to both actives and inactives? Observe that the tax-benefit system (τ, D) studied here is completely equivalent to a universal basic income $(\tau' - D, D)$ for a τ' chosen such as $y - \tau'(y) + D = y - \tau(y)$, i.e. both systems decentralize the very same allocation. In other words, it is the (consequentialist) difference between active and inactive transfers that has welfare consequences, not the (deontological) means of transfers³¹. For clarity, I retain the formulation with an inactivity benefit.

An incentive-compatible allocation (c, l) decentralized by the tax-benefit system (τ, D)

³⁰In practice, a screening mechanism may be imperfect and costly. I discuss the sensitivity of the results to this assumption in section 5.1.

³¹In layman terms, whether a job-seeker receives 1000 USD as social assistance or 500 USD as basic income combined with 500 as social assistance yield the same bundle for that jobseeker.

is such that each agent gets their preferred bundle in their budget $B(\tau, D, w_i, h_i)$:

$$\forall i, j \in \mathcal{I}, (c_i, l_i) \succsim_i (c_j, l_j) \text{ or } (c_j, l_j) \notin B(\tau, D, w_i, h_i)$$

Before turning to the main results, we need one assumption. It ensures that there exists at least one agent in the economy that has the worst endowments and *hardworking* preferences, i.e. no preference for leisure and a disutility of participation such that she is indifferent between social assistance and inactivity.

Assumption (Hardworking Poor Existence). *For any incentive-compatible allocation (c, l) decentralized by (τ, D) and any economy $e \in E$, there exists an agent $j \in \mathcal{I}$ such that $(w_j, h_j) = (\underline{w}, \underline{h})$ and $(\underline{h} + D, -1) \sim_j (-\tau(0), l_j)$ for all $l_j \in [0, 1]$.*

In other words, this Hardworking Poor agent has preferences such that she is indifferent between all the points on the frontier of the budget $B(\tau, D, \underline{w}, \underline{h})$. This assumption may seem strong for small economies but is arguably more reasonable when designing tax-benefit systems for large economies: since governments cannot observe indifference curves, it seems prudent to design policies as if any type of agents existed, should that indeed be the case in reality.³²

The next theorem translates the allocation ordering implied by the axioms from a function expressed in abstract well-being measures (in [Theorem 1](#)) to a function expressed in terms of policy tools and economy's parameters. It is stated and proven for the case where the smallest wage rate in the population is 0, i.e. $\underline{w} = 0$. It can be interpreted either as agents unable to work or as involuntary unemployed agents.³³ The case $\underline{w} > 0$ can be derived in a similar way and is relegated to [Appendix B.1](#).

Theorem 2. *Under Hardworking Poor Existence, for $e \in E$ such that $\underline{w} = 0$, and any two incentive-compatible allocations (c, l) and (c', l') decentralized by (τ, D) and (τ', D') respectively, one has that (c, l) is socially preferred to (c', l') with respect to the equality of opportunity social welfare function whenever*

$$\min \left\{ \underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w} \right\} > \min \left\{ \underline{h} + D' - \tilde{h}; -\tau'(0) - \tilde{w} \right\} \quad (1)$$

Proof. Consider the tax-benefit system (τ, D) that decentralizes (c, l) . By Hardworking Poor Existence, $\exists j \in \mathcal{I}$ with $(w_j, h_j) = (\underline{w}, \underline{h}) = (0, \underline{h})$ with preferences such that she is

³²Because Hardworking Poor Existence forces a match between preferences and the tax-benefit system under consideration, it restricts the set of tax-benefit systems that can be studied. In particular, it cannot be that $-\tau(0) < D + \underline{h}$ because there is no negative disutility of participation that could render an agent indifferent between social assistance and inactivity in such case. I argue that this is inconsequential because, in such a tax system where no one chooses social assistance $-\tau(0)$, it is completely free for the government to raise social assistance until $-\tau(0) = D + \underline{h}$.

³³Involuntary unemployment corresponds to the case in which the state of the labor market nullifies an agent's productivity.

indifferent between all bundles on the frontier of her budget. At the allocation (c, l) , that Hardworking Poor agent receives a bundle $(c_j, l_j) \in B(\tau, D, 0, \underline{h})$ and her indifference curve through (c_j, l_j) passes through each point of the frontier of the budget $B(\tau, D, 0, \underline{h})$. We illustrate these preferences in Figure 5.

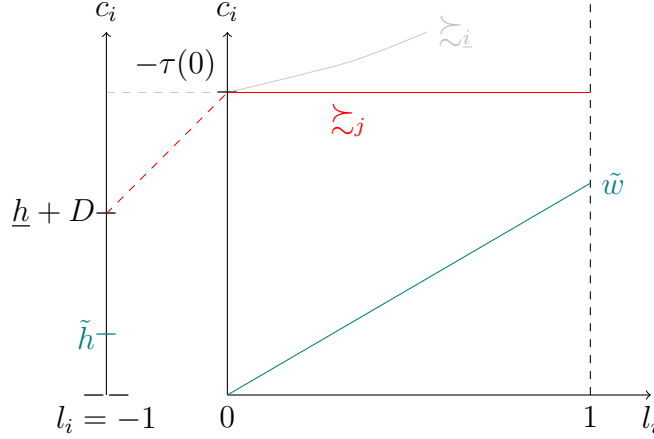


Figure 5: Illustration of the proof of Theorem 2. Hardworking Poor Existence guarantees the existence of the red indifference curve of agent j that is indifferent between social assistance and the worst consumption when inactive. Because $\underline{w} = w_j = 0$, that agent cannot do better than $-\tau(0)$ in the labor market. The welfare of agent j is the transfer that makes her indifferent to the green budget $B(0, \tilde{w}, \tilde{h})$, i.e. the smallest transfer that translates the green budget to intersect with the red indifference curve.

Using Definition 1, we can compute the money-metric welfare of that agent j . Graphically, it consists in finding the smallest transfer t that translates the green budget $B(0, \tilde{w}, \tilde{h})$ to make it intersect with the red indifference curve of agent j . Algebraically, this is

$$M_j(c_j, l_j) = \min_i t \text{ s.t. } \begin{cases} -\tau(0) = \tilde{w}\hat{l} + t & \text{if } \hat{l} \in [0, 1], \\ \underline{h} + D = \tilde{h} + t & \text{if } \hat{l} = -1 \end{cases}$$

Because of Hardworking Poor agent j 's flat indifference curve when active, the smallest value of t when $\hat{l} \in [0, 1]$ can only be found when $\hat{l} = 1$. Hence, the money-metric utility of j reduces to

$$M_j(c_j, l_j) = \min \left\{ \underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w} \right\}$$

Let me define $\underline{t} = \min \left\{ \underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w} \right\}$ and consider the first-best budget defined by $B(\underline{t}, \tilde{w}, \tilde{h})$, i.e. the smallest green reference budget which intersects with the red indifference curve in Figure 5. By construction, the consumption in $B(\underline{t}, \tilde{w}, \tilde{h})$ when inactive is $\tilde{h} + \underline{t}$ which is no larger than $\underline{h} + D$. Moreover, the consumption when working is $\tilde{w}l + \underline{t}$ which is no larger than $-\tau(0)$.

For any agent i with $(w_i, h_i) \geq (0, \underline{h})$, because preferences are increasing in consumption, $(h_i + D, -1)$ is preferred by agent i to $(\underline{h} + D, -1)$ and because preferences are also nonincreasing in labor, $(-\tau(0), 0)$ is weakly preferred by agent i to any bundle $(-\tau(0), l)$ with $l \in [0, 1]$. This proves that, in the preference ranking of i , the inactive bundle $(h_i + D, -1)$ dominates the inactive bundle of $B(\underline{t}, \tilde{w}, \tilde{h})$ while the unemployment bundle $(-\tau(0), 0)$ dominates any bundle with $l \in [0, 1]$ in $B(\underline{t}, \tilde{w}, \tilde{h})$.

Both the inactive bundle $(h_i + D, -1)$ and the unemployment bundle $(-\tau(0), 0)$ belong to the budget $B(\tau, D, w_i, h_i)$ for any i with $(w_i, h_i) \geq (0, \underline{h})$. Moreover, because (c, l) is an incentive-compatible allocation, (c_i, l_i) is weakly preferred by agent i to both $(h_i + D, -1)$ and $(-\tau(0), 0)$. The combination of the arguments yields that (c_i, l_i) is weakly preferred by agent i to any point in $B(\underline{t}, \tilde{w}, \tilde{h})$. Because $M_i(\cdot)$ represents preferences, we have that $M_i(c_i, l_i) \geq \underline{t}$ by Definition 1. Since this holds for all $i \in \mathcal{I}$, we have $\min_{i \in \mathcal{I}} M_i(c_i, l_i) = \min \left\{ \underline{h} + D - \tilde{h}; -\tau(0) - \tilde{w} \right\} = M_j(c_j, l_j)$. By Theorem 1, the result follows. ■

Theorem 2 is useful because it amounts to identifying a sufficient statistic for evaluating the desirability of any tax-benefit reform, even in a suboptimal world. It can be used to compare a reform scenario (τ, D) against the status quo tax-benefit system (τ', D') . For example, if the gap between the lowest consumption and the average production in the home sector $(\underline{h} + D - \tilde{h})$ is larger in absolute value than in the labor market $(-\tau(0) - \tilde{w})$, then a welfare-improving reform would consist in raising the inactivity benefit D . I come back to the usefulness of this theorem in the empirical section 5. Let me also note that one could be concerned that Hardworking Poor Existence is too strong because it leads to pathological preferences as represented in Figure 5. Fortunately, I show in Appendix B.1. that similar results can be obtained with much weaker assumptions: we only need to assume that the slope of the tax-benefit system is tangent to the indifference curves of some agents, which is necessarily true if there are no gaps in the income distribution.

What is the intuition behind this result? Recall that, if there were no inequalities in productive skills (w_i, h_i) , everyone would have a budget $(\tilde{w}, \tilde{h})$ and the absence of redistribution would be optimal by **Responsibility**. Hence, the government assesses welfare by computing the gap between the current situation of an individual and what she would have consumed in the ideal situation where she would have had the average budget $(\tilde{w}, \tilde{h})$. Because of the *maximin* nature of the social objective, any extra dollar should *fully* go to the agent who has the largest gap. As argued above, the social welfare function requires us to focus on redistribution among actives and among inactives, but not between them. Hence, the government computes that gap in the labor market and compares it to the gap at home. Overall, Theorem 2 implies that reducing within-sector inequality is welfare-improving in the sector that has the largest level of inequality, and

the sectoral inequality measure is simply the difference between the smallest consumption and the average productivity.

Let me now turn to the partial characterization of the optimal tax-benefit system. It is enough to concentrate on D^* for our purposes.

Theorem 3. *Under Hardworking Poor Existence, for $e \in E$ with $\underline{w} = 0$, if there exists an incentive-compatible allocation decentralized by (τ^*, D^*) that is optimal with respect to the equality of opportunity social welfare function, then*

$$D^* = \tilde{h} - \underline{h} - \tau^*(0) - \tilde{w}$$

The proof is immediate from [Theorem 2](#). The optimal inactivity benefit follows a similar formula for economies with $\underline{w} > 0$ ³⁴ and is relegated to [Appendix B.1](#).

This optimal inactivity benefit formula is interesting for several reasons. First, it gives us an idea of the consumption difference that should hold between actives and inactive. In other words, this theorem pinpoints the premium that active agents *fairly* deserve because of their participation in the labor market, if any, the magnitude of which must be pinned down by empirical moments of the economy. In particular, observe that it could be the case that $D^* = -\tau^*(0)$ and social assistance is given to both actives and inactive agents. This is optimal when the bottom and the average productivities are equal across sectors, i.e. $\underline{h} = 0 = \underline{w}$ and $\tilde{h} = \tilde{w}$. This suggests that the discontinuity in transfers between inactive and unemployed is not a built-in feature of the extensive labor supply model, but rather depends on the particular economy under consideration.

Second, this premium is increasing in the discrepancy between average productivity in the labor market and at home ($\tilde{w} - \tilde{h}$). This suggests that as the labor market becomes relatively more productive, the inactivity benefit should decrease for a given τ^* . This would come as no surprise if we had introduced a government budget constraint. What is peculiar, surprising, and new here is that this holds only because of fairness considerations, whatever the size of efficiency costs.

Third, this theorem also informs us on the complementarity/substitutability of the inactivity benefit with the traditional safety net programs. In particular, basic income should supplement, rather than crowd out, the standard safety net as they covary in identical rather than opposite directions: an increase in one dollar of optimal social assistance $-\tau^*(0)$ should be accompanied with an increase of one dollar in optimal inactivity benefit D^* , *ceteris paribus*. Intuitively, if the public finance constraint is marginally relaxed and the optimal social assistance increases by a dollar, then the gap between the worst consumption in the labor market and the average wage shrinks by a dollar.

³⁴The main difference is that the worst-off in the labor market may be different from the ones earning $-\tau(0)$. For example, they may work full time at the minimum wage.

As a result, the government increases the inactivity benefit by a dollar to shrink the gap between the lowest and average home production by the same magnitude.

Finally, while [Theorem 3](#) characterizes the relative level of the safety net and the inactivity benefit at the optimum, it is silent about the joint absolute level. This is why it is only a partial characterization of the optimal tax-and-transfer system. To sketch its design³⁵, consider a given economy e and the *Laissez-faire* policy $(\tau, D) = (0, 0)$. The government computes the gap between the worst consumption and the average production in the labor market and in the home sector using [Theorem 2](#). In most cases, that gap is much greater in the former than in the latter sector. Hence, the government will start by directing some transfers to the worst-off in the labor market, up to the point where the gaps between the worst consumption and the average production are equalized across sectors, as embodied by [Theorem 3](#). In turn, the Rawlsian government will pursue tax collection by setting taxes to the revenue-maximizing rate, i.e. the tip of the Laffer curve, and transfer all the proceeds through D^* and $-\tau^*(0)$. In other words, it will direct redistribution such that any dollar spent on D is matched with a dollar spent on the safety net, thereby setting their joint level as high as efficiency permits.

5 Empirical application

Whether introducing an inactivity benefit is a welfare-improving reform or an optimal policy is ultimately an empirical question, as one can notice from [Theorem 2](#) and [Theorem 3](#), respectively. Let me start with the analysis of welfare-improving reforms.³⁶ I therefore need to estimate parameters $-\tau(0)$, $\underline{h}$, $\tilde{w}$, $\tilde{h}$ from equation (1) in [Theorem 2](#).

I set the right-hand side of equation (1) with the status-quo tax-benefit system such that $(-\tau'(0), D') = (-\tau'(0), 0)$ for all countries as to date no OECD government has waived the conditionality of welfare benefits to labor market participation. Then, I calibrate the left-hand side with the reform that consists in giving a positive inactivity benefit, i.e. the reform $(-\tau(0), D)$ with $D > 0$. By [Theorem 2](#), this reform is welfare-improving if

$$-\tau(0) - \tilde{w} > \underline{h} - \tilde{h} + D \iff -\tau(0) > \underline{h} - \tilde{h} + D + \tilde{w}$$

The right-hand side of this inequality reads as a sufficient statistic for the desirability of the reform of the inactivity benefit. In what follows, I consider a dollar increase of D (i.e. from 0 dollar to 1 dollar) and I compute the right-hand side. I obtain a sufficient

³⁵It is easy to prove that τ^* would have the same qualitative properties as Fleurbaey and Maniquet (2007), i.e. setting $\underline{w} - \tau(\underline{w})$ as high as possible with negative marginal tax rates over $[0, \underline{w}]$.

³⁶Observed tax-benefit systems around the world are probably far away from the Mirrleesian optimum such that policy recommendations of practical use are more likely to emerge from the study of welfare-improving reforms.

statistic that answers the following question: *by how much should one increase the safety net in order for 1 dollar of inactivity benefit to be welfare-improving?* Hence, this section performs a bounding exercise.

The advantage of the sufficient statistics (1) is that estimates for the current social assistance $-\tau'(0)$ as well as the average gross earnings $\tilde{w}$ are readily available statistics for many countries³⁷. However, the difficulty of this exercise is that home production surpluses estimates for $\underline{h}$ and $\tilde{h}$ are not readily available. From [section 2](#), we know that h_i must be measured as the product of the sunk time cost of participating in the labor market³⁸ with the productivity in the home sector. Let me denote the former by F and the latter by γ_i :

$$\underbrace{h_i}_{\text{home surplus}} = \underbrace{F}_{\text{Time cost of participation}} \times \underbrace{\gamma_i}_{\text{home hourly productivity}}$$

To estimate these two key unobservables, I will take a Beckerian view on home production and set average productivities to be identical across the two sectors, i.e. $\tilde{\gamma} = \tilde{w}$ (Becker, 1965). Moreover, I impose that $\underline{\gamma} = 0$. Observe that I only impose two moments restrictions, i.e. on the minimum and the average, while staying completely agnostic about the shape of the γ_i distribution. [Section 5.1](#) addresses robustness.

To get an estimate of F , I exploit the recent G-SWA survey on time savings when working from home (Aksoy et al., 2023). On average across countries, workers spent 72 minutes per day commuting which is taken to reflect the sunk cost of labor market participation³⁹. In turn, F is expressed as the fraction of this time cost over the statutory length of the working week⁴⁰ because l is normalized to 1. For example, the average American spends 55 minutes commuting per day over a 40-hour workweek, yielding a $F_{US} = 11.46\%$.

The results are reported in [Table 1](#). The primitive inputs behind each entry (the average wage $\tilde{w}$, the social assistance $-\tau(0)$, and the commuting time and working week behind F) are reported country by country in [Appendix D](#).

³⁷I recover $-\tau'(0)$ and $\tilde{w}$ from the OECD (2020) tax-benefit simulator for 29 developed economies. The results are differentiated for two different family compositions (i.e. two different status-quo τ'): childless singles and lone parents with two children. The reference year is set to 2020 for the case $\underline{w} = 0$ because there is little doubt that there has been involuntary unemployment.

³⁸Observe that in the real world, this sunk time cost can only be partly controlled by the government (through job-seeking ordeals). This paper focuses on the redistribution problem considering that these ordeals are set for other reasons (see e.g. Rafkin et al. (2024)). Accordingly, the measurement of F is set to match commuting time, which is largely independent of these ordeals.

³⁹An alternative strategy would have been to use estimates of the time devoted to job search activities. However, Mukoyama et al. (2018) documented that unemployed Americans spend on average 31.1 minutes per day searching for a job, such that my choice is again conservative.

⁴⁰Additional details on the empirical application are relegated to [Appendix D](#).

Table 1: Summary of results. Sufficient increase in $-\tau(0)$ for $D = 1$ to be a welfare-improving reform, in percentage of current $-\tau(0)$.

Country	Lone parents	Singles
Australia	365.99%	410.22%
Belgium	142.67%	248.18%
Bulgaria	352.10%	1109.41%
Canada	353.95%	600.39%
Czech Republic	291.88%	706.75%
Estonia	116.44%	657.54%
France	121.76%	372.93%
Greece	294.24%	589.93%
Germany	218.40%	744.30%
Croatia	325.47%	793.48%
Hungary	1490.03%	1490.03%
Israel	223.35%	549.81%
Ireland	186.29%	288.84%
Japan	68.66%	319.12%
Lithuania	224.54%	788.99%
Latvia	302.19%	1307.68%
Luxembourg	82.64%	189.04%
Malta	205.47%	235.06%
Netherlands	257.60%	257.60%
New Zealand	164.09%	295.15%
Poland	188.73%	994.14%
Portugal	244.71%	589.42%
Romania	1012.90%	2697.91%
Slovenia	35.97%	245.36%
Slovak Republic	511.34%	1235.17%
Spain	220.58%	371.08%
Turkey	3639.13%	∞
United Kingdom	226.69%	638.80%
United States	762.76%	2163.63%

Despite a series of conservative assumptions, I find that for all 29 countries, the safety net should be increased by very large amounts before any dollar spent on inactivity benefit constitutes a welfare improvement. The magnitude of estimates is in general smaller for lone parents than for childless singles, as most countries typically offer more generous coverage to the former than the latter. These results come from the fact that distances to the average are much larger in the labor market than in the home sector, or equivalently, that the well-being measurement pinned down by the axioms always identifies the worst-off as being a job-seeker in these economies, but never an inactive. It is worth pointing out that, while results for the case $\underline{w} > 0$ are qualitatively similar, the magnitude is considerably smaller (see [Appendix B.2](#)). Yet, the policy conclusion is unchanged.

This analysis of welfare-improving reforms suggests that before introducing a basic income, a prioritarian government should significantly (most of the time, unrealistically) increase the safety net coverage offered to the actives. This comes from the fact that the empirical difference between inactives consuming $\underline{h}$ and unemployed consuming $-\tau(0)$ is smaller than $\tilde{h} - \tilde{w}$.

Now, recall that our previous theorems not only informed us of the direction of welfare-improving reforms but also on the design of optimal redistribution. [Theorem 2](#) gives a sufficient statistic that estimates by how much we should increase social assistance before a dollar of basic income is welfare-improving, while [Theorem 3](#) says how high basic income should be if the current social assistance is assumed to be optimal.⁴¹ Conveniently, these two exercises are numerically equivalent, in the sense that they deliver the same estimates (i.e. the estimates of [Table 1](#)). For example, if one assumes that the observed Luxembourgian social assistance programs are optimal ($-\tau'(0) = -\tau^*(0)$), then the optimal inactivity benefit for lone parents in Luxembourg should amount to -82.64% multiplied by the level of social assistance. In other words, all optimal inactivity benefits should be negative. Overall, this goes against the idea that introducing a basic income without modifying the safety net would be welfare-improving, let alone optimal.

5.1 Discussion

I made several assumptions that, if relaxed, would render the conflict between basic income and equality of opportunity even stronger. I discuss them in turn for the empirical application, the theoretical framework and the government's objective.

From the theorems of [section 4](#), we know that a positive basic income is more likely to emerge if the gap $\tilde{w} - \tilde{h}$ is small. In order for my negative result on its desirability to be robust, I have made measurement assumptions that were favorable to the emergence of a basic income, i.e. conservative assumptions. First, the ratio of the average productivity in the home sector to the labor market $\frac{\tilde{y}}{\tilde{w}}$ has been set to 1 while [Bridgman et al. \(2018\)](#) found it closer to 0.3. Second, the choice of F could have been smaller if it had reflected time devoted to job search rather than commuting (see [Mukoyama et al. \(2018\)](#)). Third, I assumed that the average marginal product of labor $\tilde{w}$ is equal to the average gross earnings. However, as firms have monopsony power, the marginal product of labor is higher than the wage⁴². All in all, these choices suggest that estimates in

⁴¹This interpretation of the results follows the line of the inverse-optimum. See [Bourguignon and Spadaro \(2012\)](#) and [Stantcheva \(2016\)](#) for an introduction and a critique of the inverse-optimum approach, respectively.

⁴²[Azar and Marinescu \(2024\)](#) and [Mas and Pallais \(2019\)](#) review the literature and consider that the marginal product of labor may be 25% larger than the average wage.

Table 1 are lower bounds.⁴³

Moreover, several features of the theoretical framework are also conservative with respect to this conflict. First, I considered that there is no negative externality of home production. However, if it includes black market activities, the government might wish to impose a Pigouvian tax on inactives and/or hold them responsible for their h_i , thereby decreasing even more the desirable level of D^* . Second, the model assumed away the existence of an intensive margin in the inactives' production function. As in Saez (2001, 2002) or Rothschild and Scheuer (2013), such an intensive margin would have driven an additional efficiency cost of raising D by disincentivizing effort in the home sector. Hence, these results would only be reinforced by including an intensive margin.

However, I have assumed that the government can perfectly distinguish a job-seeker from an inactive at zero cost, which might seem a strong assumption. The value of this screening device is the transfer it withholds: monitoring an inactive is worthwhile as long as it costs less than paying her the safety net $-\tau(0)$, since extending it to inactives would leave the well-being of the worst-off unchanged. Hence, costly monitoring does not affect the estimates of Table 1, and can only justify abandoning the screening device when the fiscal cost of monitoring an inactive exceeds the safety net itself.⁴⁴ In the same vein, if the screening mechanism is imperfect and a basic income emerges as optimal policy, then this policy would be justified on third-best considerations but not on second-best arguments (see Ghatak and Maniquet (2019) for a discussion).

Finally, this result holds despite the fact that the social objective has not favored the production in the formal sector with respect to the one in the home sector⁴⁵ even if there may be good reasons to do so. Indeed, there have been recent calls in the public debate for an increase in the employment rate, notably in order to maintain the sustainability of public pension systems in aging economies.

However, this paper did not exclude the possibility that other social objectives may lead to different policy recommendations. For example, a social welfare function aggregating the same money-metric utility $M_i(\cdot)$ but with a lower degree of inequality

⁴³In-kind benefits and housing benefits are excluded from the analysis. In countries where housing benefits are conditional on labor market participation, including them would be more favorable to the emergence of a basic income. However, symmetrically, including the childcare costs that participating lone parents incur, which are only partially covered by childcare support, would push in the opposite direction in most countries.

⁴⁴By contrast, a screening cost borne by the unemployed (such as ordeals) would reduce the consumption of the worst-off active one-for-one, thereby increasing the estimates of Table 1, which are then again lower bounds.

⁴⁵This is especially relevant to the debate because proponents of basic income have argued that one should not be paternalistic about what a good life is (see in particular Van Parijs (1995)).

aversion could lead to smaller estimates in Table 1.⁴⁶ Moreover, among the family of social objectives pursuing equality of opportunity, the one studied in this paper may be criticized by some because it holds agents responsible for their disutilities of participation. In Appendix C, I study the case where disutilities of participation are driven by a welfare recipient stigma⁴⁷ that the government wishes to compensate for. It is shown that the estimates in Table 1 can be then interpreted as lower bounds for the government's willingness to compensate for the stigma cost of conditionality.

6 Conclusion

This article has explored redistribution between active and inactive agents. The inquiry showed that abandoning the conditionality of social benefits on labor market participation is unlikely to be justified by the ethics of equality of opportunity in most developed economies.⁴⁸ Equivalently, if one wants to introduce a basic income in these economies, it can only be justified by a social welfare function that encourages redistribution across preferences.

The main explanation lies in the fact that as the value of home production is small, inactivity is driven by preferences for which agents are held responsible. Hence, an inequality-averse government can fight inequalities outside the labor force by providing better opportunities within the labor market, which is desirable under the proviso that the aggregate technology in the formal sector is productive enough.

This paper assumed perfect observability of the activity status a_i which might be noisy in practice.⁴⁹ This could justify the emergence of a basic income for third-best considerations, while the present paper is confined to the analysis of the second-best. Similarly, this paper has left exogenous the definition of the eligibility requirements to be considered as an active job-seeker. However, it is well-known that there exists a

⁴⁶A lower inequality aversion violates the combination of **Weak Transfer** and **Hansson Independence**, so one would need a different axiomatic structure to justify such a social welfare function.

⁴⁷This *welfare stigma hypothesis* has received attention from the theoretical literature (Besley & Coate, 1992a; Hupkau & Maniquet, 2018; Lindbeck et al., 1999; Moffitt, 1983) and was recently backed by experimental evidence (Friedrichsen et al., 2018).

⁴⁸A priori, this ethical standpoint could have justified both an anti- and a pro-basic income argument as could be attested by the debate between Rawls (1988) and Van Parijs (1991) on whether Malibu surfers should be fed. What the present paper has done is precisely to reconcile these two diametrically opposed interpretations of a single fairness viewpoint, by proving the conjecture outlined in Van Parijs and Vanderborght (2017, p.112): "[Once leisure is included in the index], the optimal option, by the standard of the difference principle, will crucially depend on the relative weights the index places on income and leisure, [...] and on a great many contingent empirical facts". In this paper, the relative weights on these dimensions are those of the agents themselves because of the non-paternalistic nature of the social objective. Then I quantify where and when these empirical facts are such that a basic income may not be justified: in developed economies.

⁴⁹Rafkin et al. (2024) study the correlations between eligibility and take-up for various US transfer programs. They find that take-up is concentrated among eligible individuals with the worst consumption levels. Hence, the correlation is positive but below 1.

heterogeneity in the stringency of these requirements across developed countries, whose positive and normative study is left for future research.

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A Axiomatic proofs

A.1 Axioms

In this section, I define formally each axiom used in the paper. Let me define the social ordering function (SOF) $\succsim$ for an economy $e \in E$ as a transitive ordering such that for any two allocations $(c, l), (c', l') \in X^{\mathcal{I}}$, such that we write $(c, l) \succsim (c', l')$ whenever (c, l) is socially weakly preferred to (c', l') . The strict social preference and the social indifference are denoted by $\succ$ and $\sim$, respectively.

The first axiom is the celebrated Pareto principle, which states that the planner must agree with unanimous decisions.

Axiom 1: Weak Pareto

If $(c_i, l_i) \succ_i (c'_i, l'_i)$ for all $i \in \mathcal{I}$ then $(c, l) \succ (c', l')$.

The second axiom holds agents responsible for inequalities spawned by their preferences. It consists in saying that when all agents have identical endowments, reducing lump-sum transfers inequality is desirable. In the limit, Laissez-faire is optimal in this knife-edge case where there is no inequality in endowments. It is illustrated in [Figure A.1](#).

Axiom 2: Responsibility

For every economy $e \in E$ and allocations with $(w_i, h_i) = (w_0, h_0) \forall i \in \mathcal{I}$,
If $\exists i, j \in \mathcal{I}$ and $(c, l), (c', l') \in X^{\mathcal{I}}$ with

$$\begin{aligned} (c_i, l_i) &\in \max_{\succsim_i} B(t_i, w_0, h_0) & (c'_i, l'_i) &\in \max_{\succsim_i} B(t'_i, w_0, h_0) \\ (c_j, l_j) &\in \max_{\succsim_j} B(t_j, w_0, h_0) & (c'_j, l'_j) &\in \max_{\succsim_j} B(t'_j, w_0, h_0) \end{aligned}$$

and $(c_k, l_k) = (c'_k, l'_k)$ for all $k \in \mathcal{I} \setminus \{i, j\}$ and $\exists \delta > 0$ such that $t'_i - \delta = t_i \geq t_j = t'_j + \delta$, then $(c, l) \succ (c', l')$.

The third axiom compensates agents for unequal skills (w_i, h_i) . It is illustrated in [Figure A.2](#).

Axiom 3: Weak Transfer

For every $e \in E$ and allocations $(c, l), (c', l') \in X^{\mathcal{I}}$, if $\exists i, j \in \mathcal{I}$ two agents with $\succsim_i = \succsim_j$ such that

$$l_i = l_j = l'_i = l'_j$$

and for some $\delta > 0$

$$c'_i - \delta = c_i \geq c_j = c'_j + \delta$$

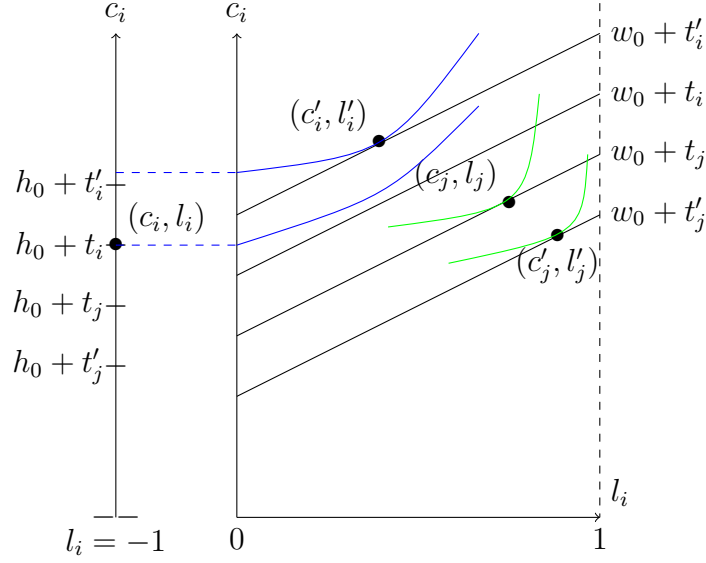


Figure A.1: In this figure, all agents have the same endowments w_i and h_i . Yet, they have received different lump-sum transfers for some reasons. **Responsibility** imposes that $(c, l) \succ (c', l')$ because lump-sum transfer inequality is reduced in (c, l) .

while $(c_k, l_k) = (c'_k, l'_k) \forall k \in \mathcal{I} \setminus \{i, j\}$;

Then, $(c, l) \succsim (c', l')$

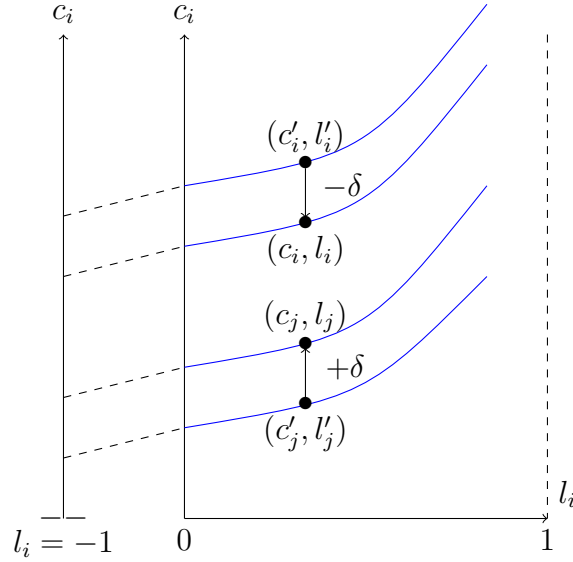


Figure A.2: **Weak Transfer** imposes that $(c, l) \succsim (c', l')$

Now, in order to build a SOF for all economies one needs consistency conditions i.e. invariance rules of the social evaluation when the economy changes. A popular choice in the literature is *Separability* which prescribes that adding or removing from the economy indifferent agents should not affect the ranking between two allocations.

Axiom: Separability

If there exist $(c, l), (c', l') \in X^{\mathcal{I}}$ such that $(c_j, l_j) = (c'_j, l'_j)$ then

$$(c, l) \succsim (c', l') \iff (c, l)_{(\mathcal{I} \setminus \{j\})} \succsim_{(\mathcal{I} \setminus \{j\})} (c', l')_{(\mathcal{I} \setminus \{j\})}$$

with $(c, l)_{(\mathcal{I} \setminus \{j\})} = \left((c_1, l_1), \dots, (c_{j-1}, l_{j-1}), (c_{j+1}, l_{j+1}), \dots, (c_I, l_I) \right)$

and $(c', l')_{(\mathcal{I} \setminus \{j\})} = \left((c'_1, l'_1), \dots, (c'_{j-1}, l'_{j-1}), (c'_{j+1}, l'_{j+1}), \dots, (c'_I, l'_I) \right)$ and $\succsim_{(\mathcal{I} \setminus \{j\})}$ is the SOF when the set of agents is $\mathcal{I} \setminus \{j\}$

We will also use a logical weakening of the latter, i.e. **Mean-Preserving Separability** which prescribes that removing indifferent agents from the economy does not affect the social ranking only if the absence of these agents does not change the per-capita amount of resources.

Axiom 4: Mean-Preserving Separability

For all $\mathcal{S} \subset \mathcal{I}$ and for every economy $e, e_{(\mathcal{I} \setminus \mathcal{S})} \in E$ with $e_{(\mathcal{I} \setminus \mathcal{S})} = \left((w_i, h_i, \succsim_i)_{\forall i \in \mathcal{I} \setminus \mathcal{S}} \right)$, and for all $(c, l), (c', l') \in X^{\mathcal{I}}$ such that

- $(c_i, l_i) = (c'_i, l'_i)$ for all $i \in \mathcal{S}$ and

- $\frac{1}{I} \sum_{i=1}^I w_i = \frac{1}{S} \sum_{i=1}^S w_i$ and $\frac{1}{I} \sum_{i=1}^I h_i = \frac{1}{S} \sum_{i=1}^S h_i$

Then $(c, l) \succsim (c', l') \iff (c, l)_{(\mathcal{I} \setminus \mathcal{S})} \succsim_{(\mathcal{I} \setminus \mathcal{S})} (c', l')_{(\mathcal{I} \setminus \mathcal{S})}$

where $(c, l)_{(\mathcal{I} \setminus \mathcal{S})}, (c', l')_{(\mathcal{I} \setminus \mathcal{S})} \in X^{\mathcal{I} \setminus \mathcal{S}}$ and $\succsim_{(\mathcal{I} \setminus \mathcal{S})}$ is the social ordering over the subpopulation $\mathcal{I} \setminus \mathcal{S}$.

Finally, the fifth axiom, **Hansson (1973) Independence**, deals with the informational structure of the SOF. It weakens the Arrovian binary independence in order to escape the impossibility of social choice. It imposes that when the indifference curves over two allocations are unchanged between two economies, then the social ordering over these allocations is unchanged as well.

Axiom 5: Hansson (1973) independence

For every economy $e = \left((w_i, h_i, \succsim_i)_{\forall i \in \mathcal{I}} \right)$ $e' = \left((w_i, h_i, \succsim'_i)_{\forall i \in \mathcal{I}} \right) \in E$ with $(\succsim_i)_{i \in \mathcal{I}}$ and $(\succsim'_i)_{i \in \mathcal{I}}$ two profiles of preferences, let $(c, l), (c', l') \in X^{\mathcal{I}}$ be two allocations,

If $\forall q \in X \quad \left[(c_i, l_i) \sim_i q \iff (c_i, l_i) \sim'_i q \text{ and } (c'_i, l'_i) \sim_i q \iff (c'_i, l'_i) \sim'_i q \quad \forall i \in \mathcal{I} \right]$

Then, $[(c, l) \succsim (c', l') \iff (c, l) \succsim' (c', l')]$

where $\succsim'$ is the SOF when the economy is e' .

I now prove that the standard axiom of *Separability* clashes with **Responsibility** and **Weak Transfer**. To the best of my knowledge, this result is new. The key difference with previous papers is that the present paper has three dimensions of heterogeneity $(\succsim_i, w_i, h_i)$ while most previous papers only had two (Fleurbaey & Maniquet, 2018).

Theorem A.1. *There is no SOF that satisfies **Responsibility**, **Weak Transfer** and **Separability** for all $e \in E$*

Proof. By contradiction, suppose the statement does not hold. Consider the economies $e_1 = \{(\zeta_i, \bar{h}, \underline{w}), (\zeta_j, \bar{h}, \underline{w})\}$, $e_2 = \{(\zeta_i, \underline{h}, \bar{w}), (\zeta_j, \underline{h}, \bar{w})\}$, $e_3 = \{(\zeta_i, \bar{h}, \underline{w}), (\zeta_j, \bar{h}, \underline{w}), (\zeta_i, \underline{h}, \bar{w}), (\zeta_j, \underline{h}, \bar{w})\}$, with $\bar{h} > \underline{h}$ and $\bar{w} > \underline{w}$, as well as their associated utility-maximizing bundles:

$$\begin{array}{ll} (c_1, l_1) \in \max_{\sim_i} B(0, \bar{h}, \underline{w}) & (c_2, l_2) \in \max_{\sim_i} B(0, \underline{h}, \bar{w}) \\ (c_3, l_3) \in \max_{\sim_j} B(0, \bar{h}, \underline{w}) & (c_4, l_4) \in \max_{\sim_j} B(0, \underline{h}, \bar{w}) \end{array}$$

The economies have been constructed such that $c_3 - c_4 = c_2 - c_1$, and $l_1 = l_2$ as well as

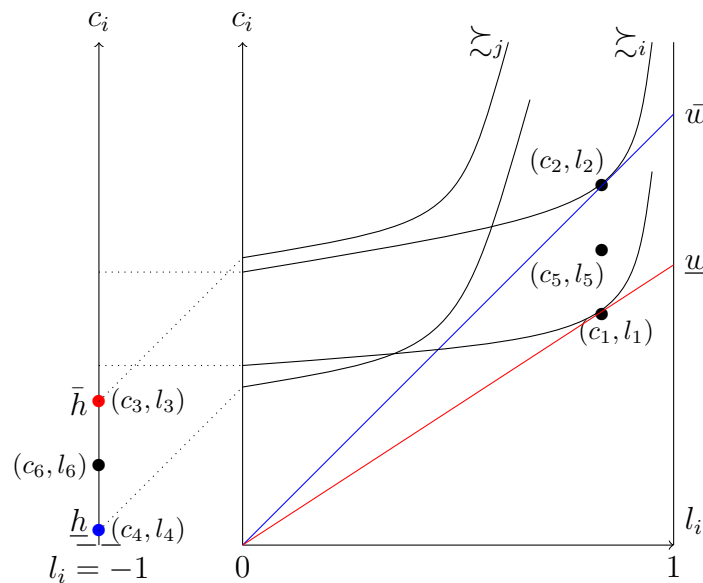


Figure A.3: Illustration of the proof of Theorem A.1.

$l_3 = l_4$. I define the average bundles $(c_5, l_5) = \frac{(c_1, l_1) + (c_2, l_2)}{2}$ and $(c_6, l_6) = \frac{(c_3, l_3) + (c_4, l_4)}{2}$ and illustrate the setup in [Figure A.3](#). It is enough to prove that the SOF leads to a contradiction in these particular economies for the statement to hold.

By **Responsibility**, $\left((c_1, l_1), (c_3, l_3) \right) \succ_{(e_1)} \left((c_5, l_5), (c_6, l_6) \right)$,

By **Separability**, $\left((c_1, l_1), (c_3, l_3), (c_2, l_2), (c_4, l_4) \right) \succ_{(e_3)} \left((c_5, l_5), (c_6, l_6), (c_2, l_2), (c_4, l_4) \right),$

By **Weak Transfer**, $\left((c_5, l_5), (c_6, l_6), (c_5, l_5), (c_6, l_6) \right) \preceq_{(e_3)} \left((c_1, l_1), (c_3, l_3), (c_2, l_2), (c_4, l_4) \right)$,

By **Transitivity**, $\left((c_5, l_5), (c_6, l_6), (c_5, l_5), (c_6, l_6) \right) \succ_{(e_3)} \left((c_5, l_5), (c_6, l_6), (c_2, l_2), (c_4, l_4) \right)$,
 By **Separability**, $\left((c_5, l_5), (c_6, l_6) \right) \succ_{(e_2)} \left((c_2, l_2), (c_4, l_4) \right)$,

But this contradicts **Responsibility**, proving the statement. ■

As argued in the main text, this paper considers that the previous impossibility should be escaped by weakening *Separability*: removing indifferent agents is not innocuous because it changes the per-capita resources, while **Responsibility** and **Weak Transfer** only compare allocations within a fixed amount of aggregate resources. *Separability* is the only axiom that bridges economies, and in the proof of Theorem A.1 it bridges economies whose per-capita endowments differ: the step from e_3 to e_1 removes the pair endowed with $(\underline{h}, \bar{w})$ and the step from e_3 to e_2 removes the pair endowed with $(\bar{h}, \underline{w})$, so that each removal makes the remaining economy richer per head in one sector and poorer in the other. **Mean-Preserving Separability** forbids precisely these two steps: it permits the removal of indifferent agents whenever it leaves per-capita endowments unchanged.

But why did per-capita endowments, or arithmetic averages emerge? Recall that **Weak Transfer** operates δ -transfers of consumption between agents supplying identical labor. Because of the linear structure of production, such transfers are isomorphic to transfers in endowments. Iterating δ -transfers drives equal-preference agents toward equal consumption, whose counterpart in endowments is the equal-endowment profile sharing the same aggregate resources. For example, in Figure A.3, (c_5, l_5) is on the budget with an hourly wage $\frac{\bar{w} + \underline{w}}{2}$. Moreover, this equal-endowment benchmark where all inequalities have been eliminated is the one that **Responsibility** privileges. Hence, the arithmetic average of endowments is the natural object that the planner should compute when it asks the key counterfactual question: how much would each agent consume if inequalities in endowments were eliminated?

The main-text Theorem 1 shows that one can escape the impossibility of Theorem A.1 by weakening **Separability** to **Mean-Preserving Separability**. I prove Theorem 1 by using two lemmas. This proof is reminiscent of Fleurbaey and Maniquet (2006, 2011) and Valletta (2014)⁵⁰. Lemma 1 implies the maximin⁵¹ nature of the social ordering while Lemma 2 characterizes the well-being measure.

⁵⁰In comparison with Fleurbaey and Maniquet (2006), I have a stronger Responsibility requirement but a weaker Separability requirement and more dimensions of heterogeneity. With respect to Valletta (2014), I have weaker versions of Pareto and Transfer axiom. That paper dealt with the fair income tax if there are two consumption goods but only one productive skill. In the present paper, I have an homogeneous consumption good but productive skills in two sectors.

⁵¹There have been recent axiomatizations of money-metric aggregator with finite inequality aversion which consists in weakening either **Weak Transfer** (Bosmans et al., 2018) or **Hansson Independence** (Piacquadio, 2017). However, I kept the present structure as the maximin has been crucial in the basic income debate.

Lemma 1. *If the SOF satisfies **Weak Pareto**, **Hansson independence** and **Weak Transfer**, then $\forall e \in E, (c, l), (c', l') \in X^I$ if there exist $\{i, j\} \in \mathcal{I}$ such that $\succsim_i = \succsim_j \equiv \succsim_0$ and*

$$(c'_i, l'_i) \succ_0 (c_i, l_i) \succ_0 (c_j, l_j) \succ_0 (c'_j, l'_j)$$

while $(c'_k, l'_k) = (c_k, l_k)$ for all $k \in \mathcal{I} \setminus \{i, j\}$, one has $(c, l) \succ (c', l')$.

Proof of Lemma 1. Follows mutatis mutandis the proof of Lemma 1 in Fleurbaey and Maniquet (2006). ■

Lemma 2. *If the SOF satisfies **Weak Pareto**, **Responsibility**, **Weak Transfer**, **Hansson Independence**, and **Mean-Preserving Separability**, and $\forall e \in E, (c, l), (c', l') \in X^I$ such that*

$$M_i(c'_i, l'_i) > M_i(c_i, l_i) > M_j(c_j, l_j) > M_j(c'_j, l'_j)$$

and $(c_k, l_k) = (c'_k, l'_k) \forall k \in \mathcal{I} \setminus \{i, j\}$

Then,

$$(c, l) \succ (c', l')$$

Proof of Lemma 2. By contradiction, suppose that $(c', l') \succsim (c, l)$.

Let me introduce two new agents, a, b such that:

- $(w_a, h_a) = (w_b, h_b) = (\tilde{w}, \tilde{h})$

- $\succsim_a = \succsim_i$ and $\succsim_b = \succsim_j$

I denote the relevant economies in the following way :

$$e_{(\{a,b\})} = \left((w_i, h_i, \succsim_i)_{\forall i \in \{a,b\}} \right) \quad e_{(\mathcal{I} \cup \{a,b\})} = \left((w_i, h_i, \succsim_i)_{\forall i \in \mathcal{I} \cup \{a,b\}} \right)$$

Let $\left((c_a, l_a), (c_b, l_b) \right), \left((c'_a, l'_a), (c'_b, l'_b) \right)$ be two allocations in $e^{\{a,b\}}$ such that

$$\begin{aligned} (c_a, l_a) &\in \max_{\succsim_a} B(t_a, \tilde{w}, \tilde{h}) & (c'_a, l'_a) &\in \max_{\succsim_a} B(t'_a, \tilde{w}, \tilde{h}) \\ (c_b, l_b) &\in \max_{\succsim_b} B(t_b, \tilde{w}, \tilde{h}) & (c'_b, l'_b) &\in \max_{\succsim_b} B(t'_b, \tilde{w}, \tilde{h}) \end{aligned}$$

with $t_a > t'_a > t'_b > t_b$

and

$$M_i(c'_i, l'_i) > M_i(c_i, l_i) > M_a(c_a, l_a) > M_a(c'_a, l'_a) > M_b(c'_b, l'_b) > M_b(c_b, l_b) > M_j(c_j, l_j) > M_j(c'_j, l'_j)$$

By Mean-Preserving Separability,

$$\left((c', l'), (c'_a, l'_a), (c'_b, l'_b) \right) \succsim_{(\mathcal{I} \cup \{a,b\})} \left((c, l), (c'_a, l'_a), (c'_b, l'_b) \right)$$

Recall that $(c_k, l_k) = (c'_k, l'_k)$ for all $k \in I \setminus \{i, j\}$. Moreover, because $M_i(\cdot)$ and $M_a(\cdot)$ represent the same preferences (resp. $M_j(\cdot)$ and $M_b(\cdot)$), we can apply Lemma 1 between agents i and a (resp. between agents j and b) to get

$$\left((c, l), (c_a, l_a), (c_b, l_b) \right) \succ_{(\mathcal{I} \cup \{a, b\})} \left((c', l'), (c'_a, l'_a), (c'_b, l'_b) \right)$$

By transitivity of the SOF,

$$\left((c, l), (c_a, l_a), (c_b, l_b) \right) \succ_{(\mathcal{I} \cup \{a, b\})} \left((c, l), (c'_a, l'_a), (c'_b, l'_b) \right)$$

By **Mean-Preserving Separability** one has,

$$\left((c_a, l_a), (c_b, l_b) \right) \succ_{(\{a, b\})} \left((c'_a, l'_a), (c'_b, l'_b) \right)$$

and this contradicts **Responsibility**, which completes the proof for lemma 2. ■

Proof of Theorem 1. The proof of theorem 1 is immediate from the combination of Lemma 1 and Lemma 2 with the characterization of the maximin ordering by Hammond (1976). □

B Results for the case $\underline{w} > 0$

B.1 Theoretical results

All the results in the main body of the paper have been presented with a worst wage rate in the economy equal to zero $\underline{w} = 0$. In this section, I show that the results can be extended to the case where $\underline{w} > 0$. This subsection derives the equivalent results as those presented in section 4 in the main text. It will not be assumed here that *Hardworking Poor Existence* holds. Rather, I will impose *Minimality* which restricts the number of tax-benefit systems (τ, D) that decentralizes a particular incentive-compatible allocation (c, l) by focusing on those where no inconsequential tax cuts are left. It formally requires that the second-best budget $B(\tau, D, \underline{w}, \underline{h})$ coincides with the intersection of lower contour sets at (c, l) for agents of type $(\underline{w}, \underline{h})$.

Assumption (Minimality). *There exists at least one agent j with $(w_j, h_j) = (\underline{w}, \underline{h})$ and the tax-benefit system (τ, D) that decentralizes the incentive-compatible (c, l) satisfies*

$$B(\tau, D, \underline{w}, \underline{h}) = \bigcap_{i: (w_i, h_i) = (\underline{w}, \underline{h})} \{(c'_i, l'_i) \in X \mid (c_i, l_i) \succsim_i (c'_i, l'_i)\}$$

When the tax-benefit system is not minimal, one can devise tax cuts that do not affect any individual nor the budget constraint of the government. It is therefore a quite natural assumption. Figure A.4 provides an example of a violation of Minimality. Although (τ, D) decentralizes (c, l) in this two-agent economy, one could find a tax cut such that

no one is affected. It would amount to make the blue locus coincide with the lower envelope of agents' indifference curves, i.e. to annihilate the shaded area in Figure A.4.

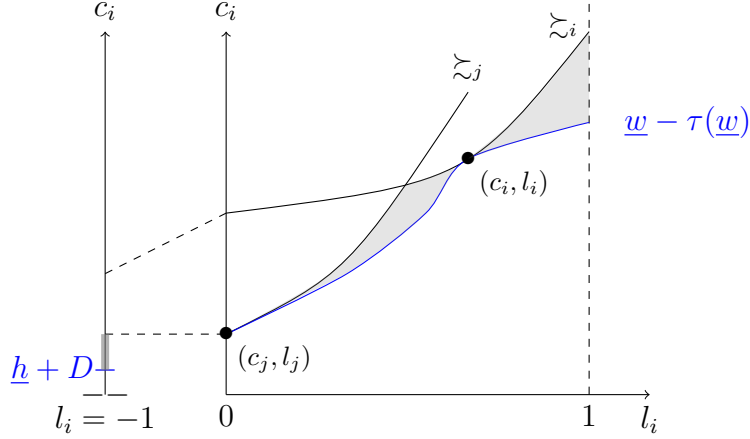


Figure A.4: Illustration of a tax-benefit system (τ, D) decentralizing (c, l) and violating Minimality. The shaded area corresponds to all points where the government could cut taxes at no cost: a tax cut (or benefit increase) in this region would expand the blue budget set $B(\tau, D, \underline{w}, \underline{h})$ but neither j nor i would change their labor supply. Formally, the shaded area is the gap between the blue budget set and the intersection of the lower contour sets of agents at (c, l) .

It is worth noting that when (τ, D) satisfies *Minimality*, $y - \tau(y)$ is non-decreasing in y on low incomes $y \in [0, \underline{w}]$ because preferences are monotonic. As a consequence, *Minimality* forbids $\frac{\partial \tau(y)}{\partial y} > 1$ on some $y \in [0, \underline{w}]$, i.e. a confiscatory tax rate on some interval of low incomes. The next theorem is reminiscent of [Theorem 2](#) in the main text. **Theorem A.2.** *Under Minimality, for $e \in E$ such that $\underline{w} > 0$, and any two incentive-compatible allocations (c, l) and (c', l') decentralized by minimal (τ, D) and (τ', D') respectively, one has that (c, l) is socially preferred to (c', l') with respect to the equality of opportunity social welfare function whenever*

$$\min \left\{ \underline{h} + D - \tilde{h}; \min_{0 \leq y \leq \underline{w}} \left(1 - \frac{\tilde{w}}{\underline{w}}\right)y - \tau(y) \right\} > \min \left\{ \underline{h} + D' - \tilde{h}; \min_{0 \leq y \leq \underline{w}} \left(1 - \frac{\tilde{w}}{\underline{w}}\right)y - \tau'(y) \right\} \quad (2)$$

Proof. Consider (τ, D) decentralizing (c, l) . By Minimality, the frontier of $B(\tau, D, \underline{w}, \underline{h})$ coincides with the lower envelope of the indifference curves of agents with endowments $(\underline{w}, \underline{h})$. Hence, the lowest money-metric welfare among agents with endowments $(\underline{w}, \underline{h})$ is obtained by finding the smallest vertical translation of the reference budget $B(0, \tilde{w}, \tilde{h})$ that intersects the frontier of $B(\tau, D, \underline{w}, \underline{h})$, that is

$$\min_{j: (w_j, h_j) = (\underline{w}, \underline{h})} M_j(c_j, l_j) = \min_i t \text{ s.t. } \begin{cases} \underline{w}\hat{l} - \tau(\underline{w}\hat{l}) = \tilde{w}\hat{l} + t & \text{if } \hat{l} \in [0, 1], \\ \underline{h} + D = \tilde{h} + t & \text{if } \hat{l} = -1 \end{cases}$$

This is equivalent to

$$\min_{j:(w_j, h_j)=(\underline{w}, \underline{h})} M_j(c_j, l_j) = \min \left\{ \underline{h} + D - \tilde{h}; \min_{0 \leq y \leq \underline{w}} \left(1 - \frac{\tilde{w}}{\underline{w}}\right)y - \tau(y) \right\}$$

It remains to show that no agent with higher endowments has lower money-metric welfare. Any agent i with $w_i \geq \underline{w}$ can obtain a given gross income $\underline{w}\hat{l} = y \in [0, \underline{w}]$ by choosing labor $y/w_i \leq \hat{l}$ and can therefore obtain the same consumption with weakly less labor than an agent of type $\underline{w}$. Similarly, since $h_i \geq \underline{h}$, the inactive bundle $(\underline{h} + D, -1)$ is weakly dominated by $(h_i + D, -1)$. Hence every bundle in $B(\tau, D, \underline{w}, \underline{h})$ is weakly dominated for agent i by some feasible bundles in $B(\tau, D, w_i, h_i)$. By incentive compatibility, agent i weakly prefers (c_i, l_i) to any bundle in $B(\tau, D, w_i, h_i)$, and therefore also to every bundle in $B(\tau, D, \underline{w}, \underline{h})$. Since this reasoning holds for all i , using the same arguments as in [Theorem 2](#) *mutatis mutandis*, it follows that $\min_{i \in I} M_i(c_i, l_i) = \min \left\{ \underline{h} + D - \tilde{h}; \min_{0 \leq y \leq \underline{w}} \left(1 - \frac{\tilde{w}}{\underline{w}}\right)y - \tau(y) \right\}$. ■

The main difference with the main-text [Theorem 2](#) is that now the worst-off in the labor market may be different from those earning the social assistance. In particular, they could be working with low pay, such as full time minimum wage earners.

Let me note that Minimality is less extreme than Hardworking Poor Existence as it does not require the existence of pathological preferences, but only imposes a mapping between the envelope of indifference curves and the tax-benefit system. If one assumes that there exists an agent that maximizes at each y such that there is no gap in the income distribution, Minimality is satisfied. Fortunately, the main text results for the case $\underline{w} = 0$ can be derived with Minimality rather than Hardworking Poor Existence by taking the limit of the reasoning when $\underline{w} \rightarrow 0$. I kept Hardworking Poor Existence for expositional purposes.

We conclude this subsection by deriving the optimal inactivity benefit with a formula reminiscent of [Theorem 3](#).

Theorem A.3. *Under Minimality, for $e \in E$ with $\underline{w} > 0$, if there exists an incentive-compatible allocation decentralized by (τ^*, D^*) that is optimal with respect to the equality of opportunity social welfare function, then*

$$D^* = \tilde{h} - \underline{h} + \min_{0 \leq y \leq \underline{w}} \left(1 - \frac{\tilde{w}}{\underline{w}}\right)y - \tau^*(y)$$

To sketch the optimal tax system, consider again the Laissez-faire policies $(\tau, D) = (0, 0)$. The government computes the smallest well-being in each sector using [Theorem A.2](#). If it sets marginal tax rates such that $\frac{\partial \tau(y)}{\partial y} = 1 - \frac{\tilde{w}}{\underline{w}}$ for all $y \in (0, \underline{w})$, i.e. negative marginal tax rates on low incomes, then all low-incomes have identical well-being. Then, the government collects taxes as much as efficiency permits and sets the relative level of D^* and $-\tau^*(0)$ as in [Theorem A.3](#). Again, this is very close to the optimal tax schedule

of Fleurbaey and Maniquet (2007), where the present paper adds the key difference between D^* and $-\tau^*(0)$ as well as the home sector.

B.2 Empirical results

This subsection extends the results of [section 5](#) in the main text to the case where $\underline{w} > 0$. I have assumed that the worst-off active is a minimum wage earner working full time, i.e. $\underline{w} = \arg \min_{0 \leq y \leq \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau^*(y)$ for all countries. Apart from this, the simulation is run for 2019 with the exact same parameters as in the main text. The results are reported in [Table A.1](#).

Table A.1: Summary of results. Sufficient increase in $\underline{w} - \tau(\underline{w})$ for $D = 1$ to be a welfare-improving reform, in percentage of current $\underline{w} - \tau(\underline{w})$.

Country	Lone parents	Singles
Australia	102.75%	116.50%
Belgium	50.47%	112.84%
Bulgaria	74.41%	139.96%
Canada	118.73%	165.95%
Czech Republic	91.40%	140.38%
Estonia	53.97%	114.82%
France	47.34%	92.28%
Greece	158.22%	158.22%
Germany	132.58%	214.20%
Croatia	129.42%	129.42%
Hungary	137.39%	233.21%
Israel	52.77%	119.34%
Ireland	46.10%	111.43%
Japan	54.23%	179.36%
Lithuania	88.95%	157.29%
Latvia	43.39%	139.87%
Luxembourg	50.49%	110.96%
Malta	93.07%	138.38%
Netherlands	116.04%	142.86%
New Zealand	44.68%	59.68%
Poland	125.05%	165.15%
Portugal	107.27%	107.27%
Romania	185.09%	192.31%
Slovenia	11.61%	97.30%
Slovak Republic	61.68%	98.93%
Spain	88.17%	100.93%
Turkey	119.83%	126.20%
United Kingdom	83.42%	134.01%
United States	103.51%	289.04%

I conclude that the results are qualitatively similar even if quantitatively different.

The policy conclusion is unchanged: governments should first increase support for low-income workers before any dollar of inactivity benefit is welfare-improving.

C Welfare recipient stigma

In this section, I consider the case of a government that does not wish to hold agents fully responsible for their preferences because their disutilities of participation have been partly driven by the *welfare recipient stigma*. It captures the idea that some agents remain inactive and do not take up conditional social benefits because enduring the screening device of the government entails a mental burden, rooted in the stigma that societies attach to benefits recipients.

Consider the following structure for the disutility of participation d_i :

$$d_i(c_i) \equiv u_i(c_i, -1) - u_i(c_i, 0) = s_i + \delta_i(c_i)$$

where $\delta_i \in [-s_i, +\infty)$ is the idiosyncratic taste parameter⁵² and $s_i \geq 0$ is the value of stigma. Observe that s_i has a money-metric interpretation: it is the maximal amount of consumption that an agent is willing to forgo in order to escape enduring the screening device of the government.

For clarity of the exposition, let me assume that there are only two draws of this stigma utility cost: $s_i \in \{S, 0\}$ with $S > 0$. Hence, there are only two different exposures to the stigma cost of conditionality in the population: those that do suffer from it and those that do not⁵³. The social planner wishes to compensate for s_i as well as w_i and h_i , while holding responsible for δ_i and their willingness to work.

Rather than deriving rigorously the full axiomatization that this compensation for s_i would entail, I sketch the result by an example. Consider the case of two fraternal twin sisters, agents k and j . They are identical in every respect but they differ in their exposure to the stigma utility cost: agent k suffers from it such that $s_k = S$ while agent j does not and $s_j = 0$.

Consider the bundle (c_j, l_j) and (c'_j, l'_j) that are such that agent j is indifferent between the two, and they are consumed when inactive and unemployed, respectively. I sketch this setup in [Figure A.5](#).

In red are drawn the indifference curves of agent k . Obviously, when she is unem-

⁵²The fact that δ_i can now have negative values reflect the possibility for some agents to derive non-pecuniary benefits from labor market participation (such as friendliness from colleagues for example) which could partly offset the burden S puts on them.

⁵³If this partitioning in two sets is degenerate, we are back to the analysis of the previous sections as there is no inequality to compensate for. In other words, if all agents experience the same stigma cost of conditionality, or if none of them does, the main formulas are unaffected.

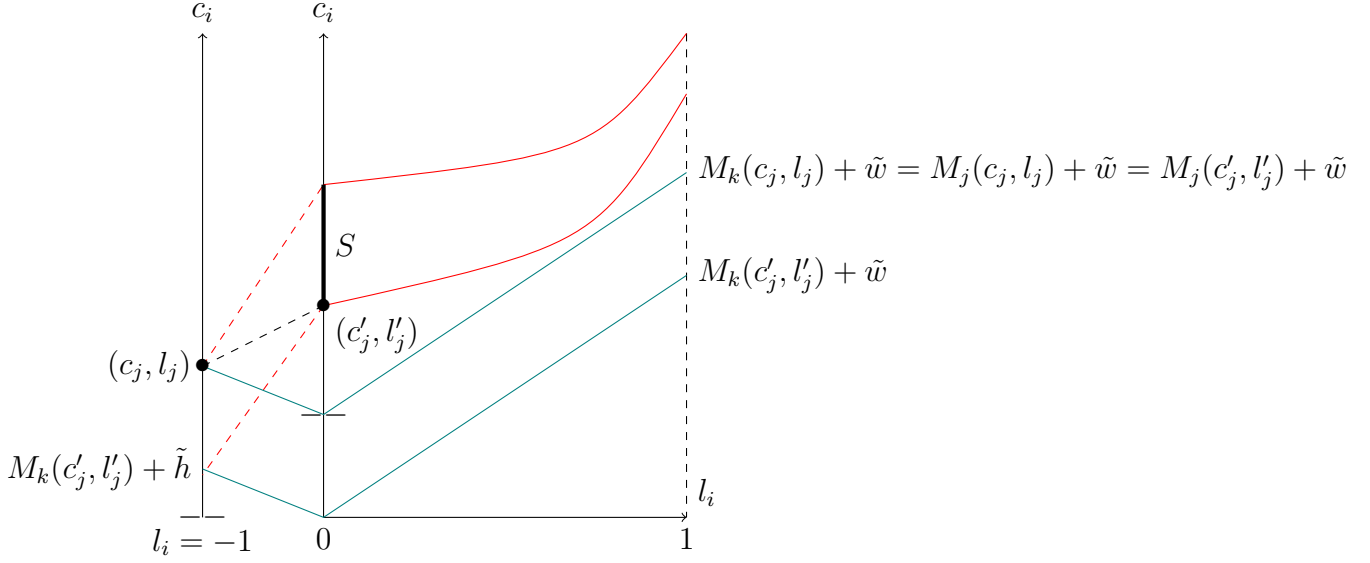


Figure A.5: Agents k and j are identical in every primitive but one: the inactive k suffers a stigma utility cost of S while the unemployed j does not. In red, the indifference curves of k . Agent j is indifferent between (c_j, l_j) and (c'_j, l'_j) .

employed and consumes (c'_j, l'_j) she suffers the stigma $s_k = S > 0$ associated with this labor market status. If one computes the $M_i(c_i, l_i)$ utility of these two agents when they consume (c'_j, l'_j) , one gets that $M_k(c'_j, l'_j) = M_j(c'_j, l'_j) - S$. In other words, *when unemployed*, the well-being measure derived in previous sections already accounts for the fact that the agent suffering from a larger disutility of participation (here, coming from the stigma) has a lower well-being.

Now, the difference is when the agents are inactive and consume (c_j, l_j) . In this case, the indifference curve of k lies above the one of j , and they only coincide at (c_j, l_j) . If one applies the $M_i(c_i, l_i)$ utility to this situation, one gets that $M_k(c_j, l_j) = M_j(c_j, l_j)$, i.e. both agents have the same level of well-being, even if agent k *would* have suffered from the stigma *had she joined the labor market*.

A government that compensates for S should treat the well-being of k when inactive as if she had actually experienced this stigma cost. In other words, the well-being measure when inactive must be reduced by s_i with respect to the $M_i(c_i, l_i)$ utility. Hence, the new well-being measure compensating for the stigma is given by

$$W_i(c_i, l_i) = M_i(c_i, l_i) - (1 - a_i)s_i$$

In turn, all the analysis above can be repeated using $W_i(c_i, l_i)$ instead of $M_i(c_i, l_i)$. Obviously, the worst-off in the home sector will now have a well-being measure of $\underline{h} -$

$S + D - \tilde{h}$. Hence, the optimal (τ^{**}, D^{**}) which maximizes $W_i(c_i, l_i)$ follows:

$$D^{**} = S + \tilde{h} - \underline{h} + \min_{0 \leq y \leq \underline{w}} (1 - \frac{\tilde{w}}{\underline{w}})y - \tau^{**}(y)$$

I conclude that S positively influences the optimal inactivity benefit in an additive fashion and thereby weakens the conflict between fairness and basic income outlined in previous sections. Because of the formalization, S has a money-metric interpretation and should be measured as an answer to this question: *how much would one be willing to pay (i.e. forgo consumption) to escape enduring the screening device of the government?*

To the best of my knowledge, such empirical estimates for S are not available in the literature. In order to justify that one dollar of inactivity benefit is welfare-improving, S must be at least as large as the estimates of [Table 1](#) converted into currency units. In many instances, it seems unrealistically large. Alternatively, S could also be interpreted as an ethical parameter. In that case, estimates from [Table 1](#) would provide lower bounds on the government's willingness to pay to escape its own screening device.

Finally, let me note that the decomposition of disutility of participation into a parameter of compensation S and a parameter of responsibility δ_i may be relevant beyond the case of stigma. For example, some empirical disutilities of participation may be driven by circumstances the government would like to compensate for, such as transportation constraints or having a disabled child. In that case, the sufficient statistics are amended as in the stigma case, and one must find the relevant value of circumstances-driven non-participation S in one's empirical context.

D Details on the empirical application

I used the OECD tax-benefit simulator TaxBEN version 2.5.0. with the following parameters:

- Childless single 2020: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning social assistance, for the year 2020. Eligible for social assistance, net of income tax and social security contributions.
- Lone parents 2020: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning social assistance, for the year 2020. Children aged 4 and 6. Eligible for social assistance, lone parents support, net of income tax and social security contributions.
- Childless single 2019: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning statutory minimum wage. Eligible for social assistance, in-work benefits, net of income tax and social security contributions.

- Lone parents 2019: aged 40, unemployed for 6 months, with 216 months of social security contributions accumulated over the lifetime, earning statutory minimum wage. Children aged 4 and 6. Eligible for social assistance, in-work benefits, lone parents support, net of income tax and social security contributions

Housing benefits are excluded because in most countries they are means-tested on income without being conditional on labor market participation, such that their inclusion would leave the sufficient statistic unaffected. More generally, in-kind benefits are outside the scope of TaxBEN, while near-cash benefits are included: for instance, SNAP constitutes the social assistance of American childless singles.

The set of countries covered by the G-SWA survey (Aksoy et al., 2023) is a strict subset of the 29 countries I study. Whenever the commuting time was not available in the G-SWA, I imputed 100 minutes per day, the largest value among the high-income countries covered by the survey (attained in Japan).

The legal length of the working week is taken from the OECD (2021) (Annex Table 5.A.1.). The statutory length may differ from the negotiated length in some countries. When both are present, I took the maximum among the two. If both are absent, I set the length of the working week to 45 hours, corresponding to the maximum of the series. I considered that the working week lasts 5 days.

Tables A.2 and A.3 report the primitive inputs behind the sufficient statistics of Table 1 and Table A.1, respectively, together with the resulting estimates. Four country-level moments fully determine each entry: the average gross annual earnings $\tilde{w}$, the daily commuting time and the length of the working week, which jointly pin down the participation time-cost parameter F , and the net income of the worst-off active for the relevant demographic group and year, namely social assistance $-\tau(0)$ in 2020 and full-time net minimum-wage earnings $\underline{w} - \tau(\underline{w})$ in 2019. Since the exercise sets $\underline{h} = 0$ and $\tilde{\gamma} = \tilde{w}$, the Beckerian estimate of the average home surplus is $\tilde{h} = F\tilde{w}$, so that a marginal inactivity benefit is a welfare-improving reform if the net income of the worst-off active exceeds the threshold $(1 - F)\tilde{w}$. Each entry of Table 1 is therefore computed as

$$\frac{(1 - F)\tilde{w} - (-\tau(0))}{-\tau(0)} = \frac{1 - F}{-\tau(0)/\tilde{w}} - 1$$

and each entry of Table A.1 replaces $-\tau(0)$ with $\underline{w} - \tau(\underline{w})$. Observe that these numbers are homogeneous of degree zero in national currency units, so inputs in national currency units do not need any purchasing-power conversion. The quantitative mechanism is then transparent: since F ranges from 11% to 22% across countries, the threshold $(1 - F)\tilde{w}$ lies between 78% and 89% of the average wage, while the net income floors range between 0% and 58% of the average wage in 2020 and between 23% and 71% in 2019.

Table A.2: Inputs and results of the empirical application, year 2020 (case $\underline{w} = 0$).

Country	$\tilde{w}$	Commute	Week	Lone parents		Singles	
				$\frac{-\tau(0)}{\tilde{w}}$	Table 1	$\frac{-\tau(0)}{\tilde{w}}$	Table 1
Australia	90 866	78	38	17.79%	365.99%	16.25%	410.22%
Belgium	50 312	<i>100</i>	38	32.17%	142.67%	22.42%	248.18%
Bulgaria	16 667	<i>100</i>	45	18.02%	352.10%	6.74%	1109.41%
Canada	76 100	65	40	19.05%	353.95%	12.34%	600.39%
Czech Republic	416 997	<i>100</i>	40	20.20%	291.88%	9.81%	706.75%
Estonia	17 224	<i>100</i>	40	36.58%	116.44%	10.45%	657.54%
France	38 002	62	35.6	38.55%	121.76%	18.08%	372.93%
Greece	18 834	58	40	22.30%	294.24%	12.74%	589.93%
Germany	51 000	65	38.2	26.95%	218.40%	10.16%	744.30%
Croatia	105 268	<i>100</i>	45	19.15%	325.47%	9.12%	793.48%
Hungary	5 043 851	66	40	5.42%	1490.03%	5.42%	1490.03%
Israel	161 172	<i>100</i>	42	24.79%	223.35%	12.34%	549.81%
Ireland	49 878	<i>100</i>	45	28.46%	186.29%	20.96%	288.84%
Japan	5 082 722	100	40	46.94%	68.66%	18.89%	319.12%
Lithuania	16 844	<i>100</i>	40	24.39%	224.54%	8.91%	788.99%
Latvia	13 656	<i>100</i>	40	19.68%	302.19%	5.62%	1307.68%
Luxembourg	64 424	<i>100</i>	45	44.61%	82.64%	28.19%	189.04%
Malta	25 902	<i>100</i>	45	26.67%	205.47%	24.32%	235.06%
Netherlands	54 510	77	37.4	23.17%	257.60%	23.17%	257.60%
New Zealand	65 079	<i>100</i>	40	29.98%	164.09%	20.03%	295.15%
Poland	61 839	54	40	30.74%	188.73%	8.11%	994.14%
Portugal	19 959	<i>100</i>	40	22.97%	244.71%	11.48%	589.42%
Romania	58 512	<i>100</i>	45	7.32%	1012.90%	2.91%	2697.91%
Slovenia	21 054	<i>100</i>	40	58.22%	35.97%	22.92%	245.36%
Slovak Republic	13 418	<i>100</i>	40	12.95%	511.34%	5.93%	1235.17%
Spain	26 028	63	40	27.10%	220.58%	18.44%	371.08%
Turkey	72 933	69	45	2.33%	3639.13%	0.00%	∞
United Kingdom	42 020	73	45	26.47%	226.69%	11.71%	638.80%
United States	59 517	55	40	10.26%	762.76%	3.91%	2163.63%

Notes: This table reports the primitive inputs behind Table 1 together with the resulting sufficient statistics, for the year 2020 and the case $\underline{w} = 0$. The column $\tilde{w}$ reports the average gross annual full-time wage, in national currency units, while $-\tau(0)$ is the annual net income of a 40-year-old out-of-work claimant of social assistance. Both moments are computed with OECD (2020) TaxBEN version 2.5.0. The “Commute” column reports the daily commuting time in minutes, taken from Table 1 in Aksoy et al. (2023), and *italics* flag the sixteen countries not covered by this survey, whose commute is imputed at 100 minutes per day, the maximum among covered countries. The “Week” column reports the workweek length in hours, the maximum of the statutory and negotiated normal weekly hours reported in Annex Table 5.A.1 of OECD (2021); **bold** flags countries absent from that table or with neither a statutory nor a negotiated limit, whose working week is imputed at 45 hours, the maximum of the series. The time cost of labor market participation is $F = (5 \times \text{Commute}) / (60 \times \text{Week})$ and the Beckerian estimate of the average home surplus is $\tilde{h} = F\tilde{w}$. The last column of each panel reproduces Table 1: a marginal inactivity benefit is a welfare-improving reform if and only if $-\tau(0)$ exceeds the threshold $\tilde{w} - \tilde{h} = (1 - F)\tilde{w}$, so that each entry equals $\frac{1-F}{-\tau(0)/\tilde{w}} - 1$, i.e. the sufficient increase in $-\tau(0)$, in percentage of its current level, for one currency unit of inactivity benefit to be welfare-improving.

Table A.3: Inputs and results of the empirical application, year 2019 (case $\underline{w} > 0$).

Country	$\tilde{w}$	Commute	Week	Lone parents		Singles	
				$\frac{\underline{w}-\tau(\underline{w})}{\tilde{w}}$	Table A.1	$\frac{\underline{w}-\tau(\underline{w})}{\tilde{w}}$	Table A.1
Australia	87 769	78	38	40.89%	102.75%	38.29%	116.50%
Belgium	49 783	<i>100</i>	38	51.88%	50.47%	36.68%	112.84%
Bulgaria	15 357	<i>100</i>	45	46.72%	74.41%	33.96%	139.96%
Canada	74 037	65	40	39.53%	118.73%	32.51%	165.95%
Czech Republic	410 579	<i>100</i>	40	41.36%	91.40%	32.93%	140.38%
Estonia	16 817	<i>100</i>	40	51.42%	53.97%	36.85%	114.82%
France	39 043	62	35.6	58.02%	47.34%	44.46%	92.28%
Greece	20 243	58	40	34.05%	158.22%	34.05%	158.22%
Germany	51 800	65	38.2	36.90%	132.58%	27.31%	214.20%
Croatia	101 362	<i>100</i>	45	35.52%	129.42%	35.52%	129.42%
Hungary	4 593 599	66	40	36.33%	137.39%	25.88%	233.21%
Israel	155 904	<i>100</i>	42	52.47%	52.77%	36.55%	119.34%
Ireland	48 858	<i>100</i>	45	55.77%	46.10%	38.54%	111.43%
Japan	5 221 760	100	40	51.33%	54.23%	28.34%	179.36%
Lithuania	15 435	<i>100</i>	40	41.90%	88.95%	30.77%	157.29%
Latvia	12 804	<i>100</i>	40	55.21%	43.39%	33.00%	139.87%
Luxembourg	60 896	<i>100</i>	45	54.14%	50.49%	38.62%	110.96%
Malta	25 069	<i>100</i>	45	42.20%	93.07%	34.18%	138.38%
Netherlands	52 970	77	37.4	38.35%	116.04%	34.11%	142.86%
New Zealand	63 255	<i>100</i>	40	54.72%	44.68%	49.58%	59.68%
Poland	58 586	54	40	39.44%	125.05%	33.47%	165.15%
Portugal	19 573	<i>100</i>	40	38.20%	107.27%	38.20%	107.27%
Romania	54 384	<i>100</i>	45	28.58%	185.09%	27.88%	192.31%
Slovenia	20 265	<i>100</i>	40	70.93%	11.61%	40.13%	97.30%
Slovak Republic	13 154	<i>100</i>	40	48.97%	61.68%	39.80%	98.93%
Spain	27 292	63	40	46.17%	88.17%	43.24%	100.93%
Turkey	61 841	69	45	39.68%	119.83%	38.56%	126.20%
United Kingdom	40 990	73	45	47.15%	83.42%	36.96%	134.01%
United States	56 577	55	40	43.51%	103.51%	22.76%	289.04%

Notes: This table reports the primitive inputs behind Table A.1 together with the resulting sufficient statistics, for the year 2019 and the case $\underline{w} > 0$. The column $\tilde{w}$ reports the average gross annual full-time wage in national currency units, while $\underline{w} - \tau(\underline{w})$ is the annual net income of a 40-year-old full-time employee earning the statutory minimum wage. Both moments are computed with the OECD (2020) TaxBEN, version 2.5.0. The “Commute” column reports the daily commuting time in minutes, taken from Table 1 in Aksoy et al. (2023), and *italics* flag the sixteen countries not covered by the G-SWA, whose commute is imputed at 100 minutes per day, the maximum among covered countries. The “Week” column reports the workweek length in hours, the maximum of the statutory and negotiated normal weekly hours reported in Annex Table 5.A.1 of OECD (2021); **bold** flags countries absent from that table or with neither a statutory nor a negotiated limit, whose working week is imputed at 45 hours, the maximum of the series. The time cost of labor market participation is $F = (5 \times \text{Commute}) / (60 \times \text{Week})$ and the Beckerian estimate of the average home surplus is $\tilde{h} = F\tilde{w}$. The last column of each panel reproduces Table A.1: each entry equals $\frac{1-F}{(\underline{w}-\tau(\underline{w}))/\tilde{w}} - 1$, i.e. the sufficient increase in $\underline{w} - \tau(\underline{w})$, in percentage of its current level, for one unit of currency of inactivity benefit to be welfare-improving.

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